

# **LAB3**

## **FOREARM**

**Dr.Ahmed AL-Baghdadi**

# **SUBJECT WILL BE DEALT**

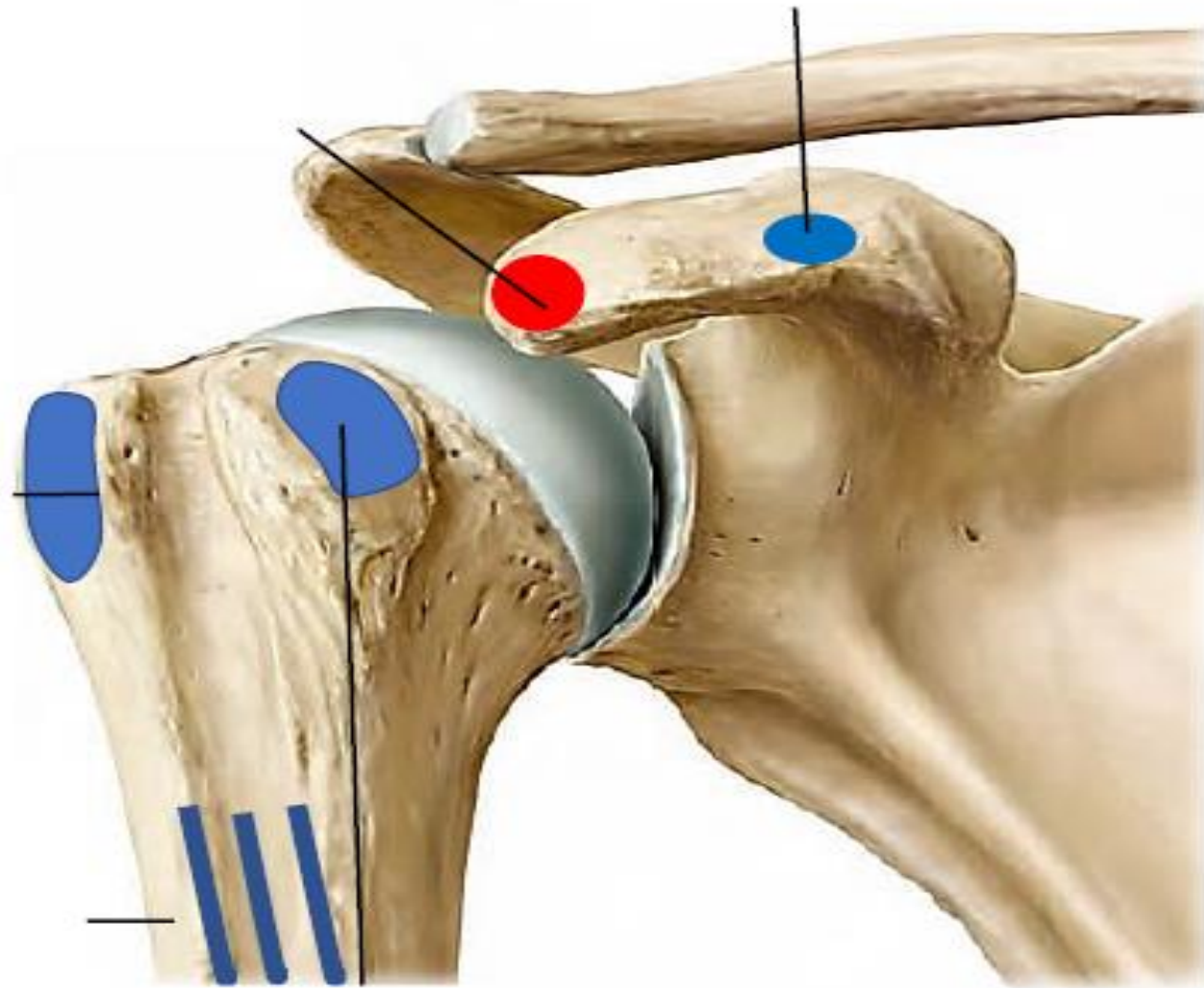
☐ **REVIEW**

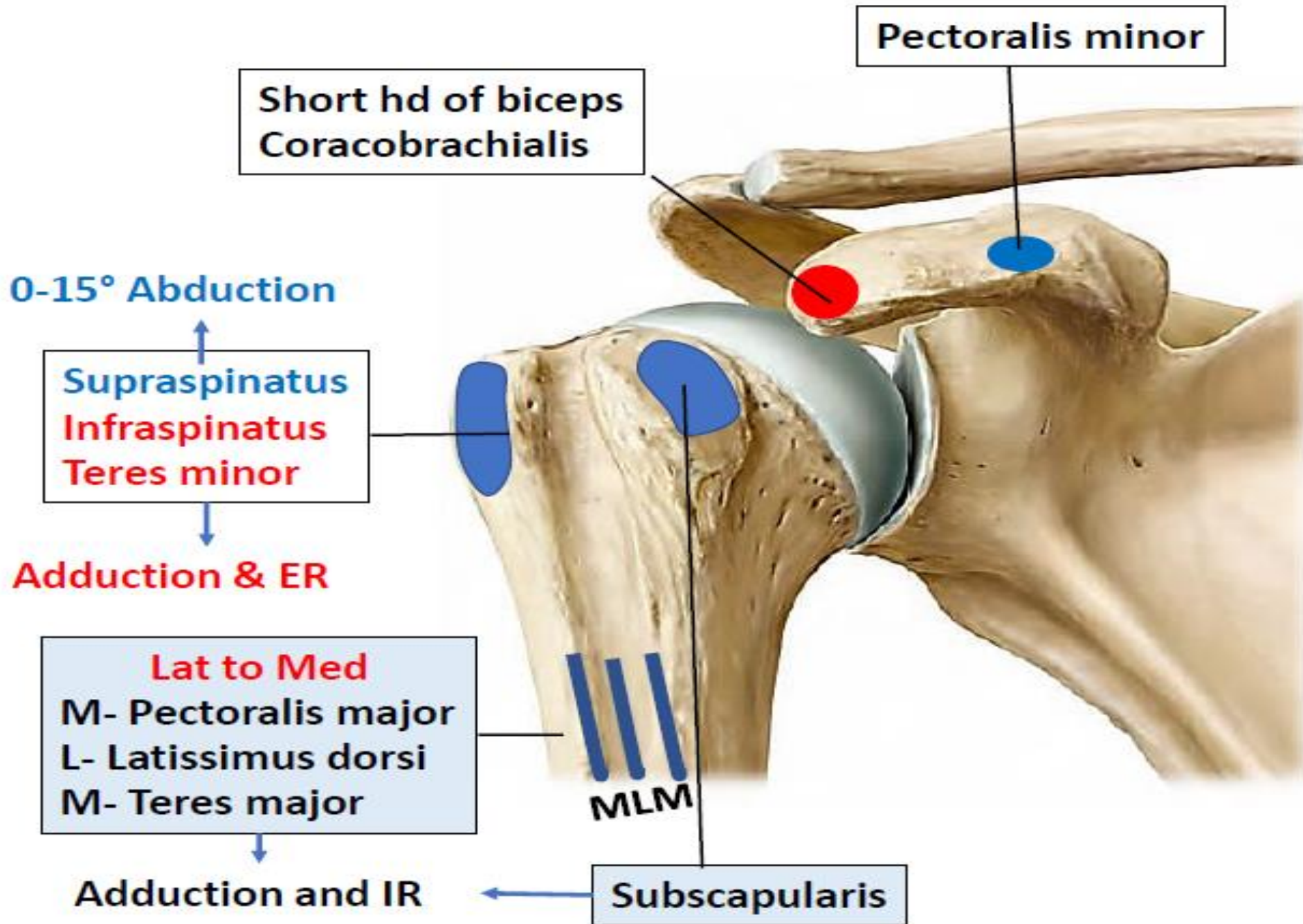
☐ **BONES**

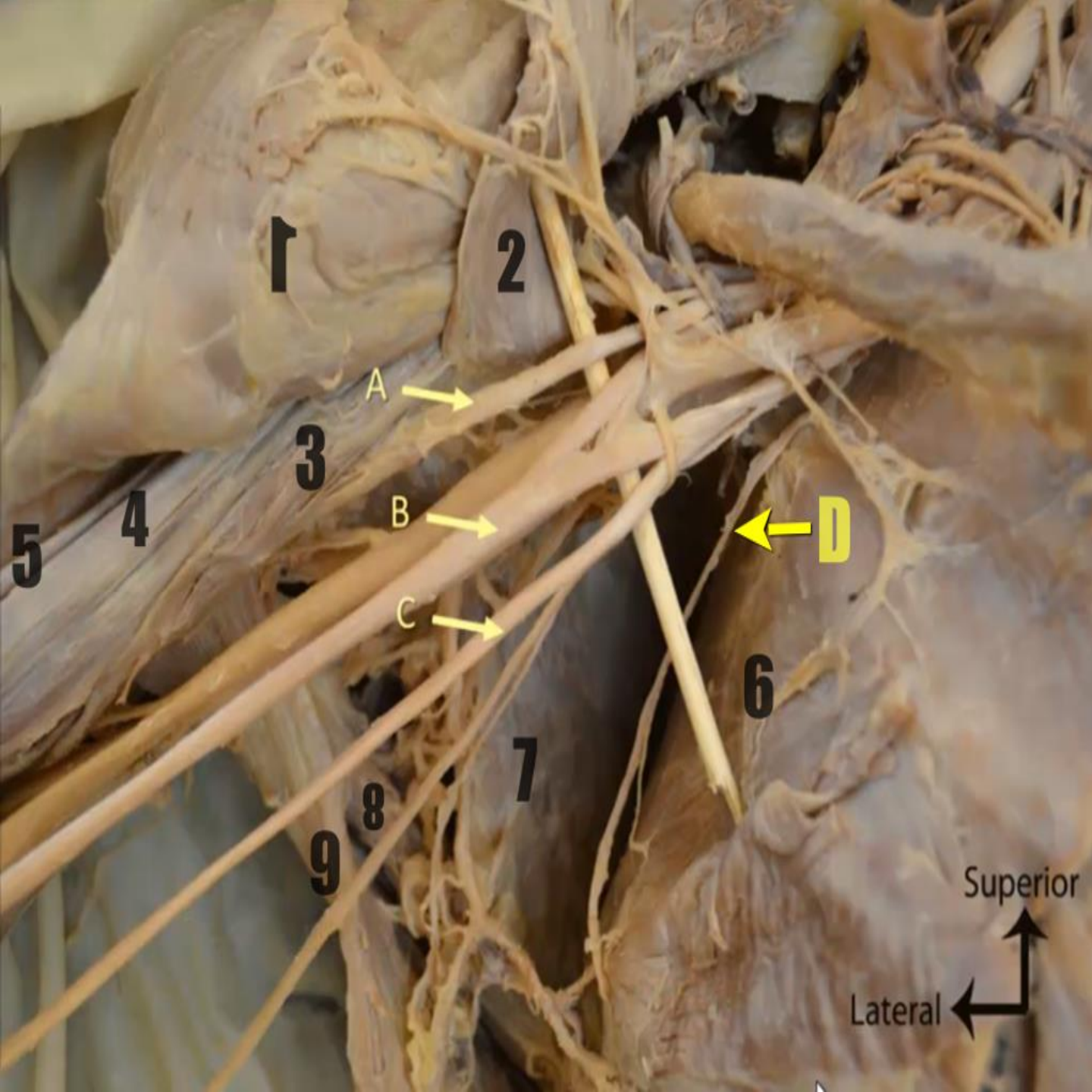
☐ **MUSCLES OF FOREARM**

☐ **VESSELES**

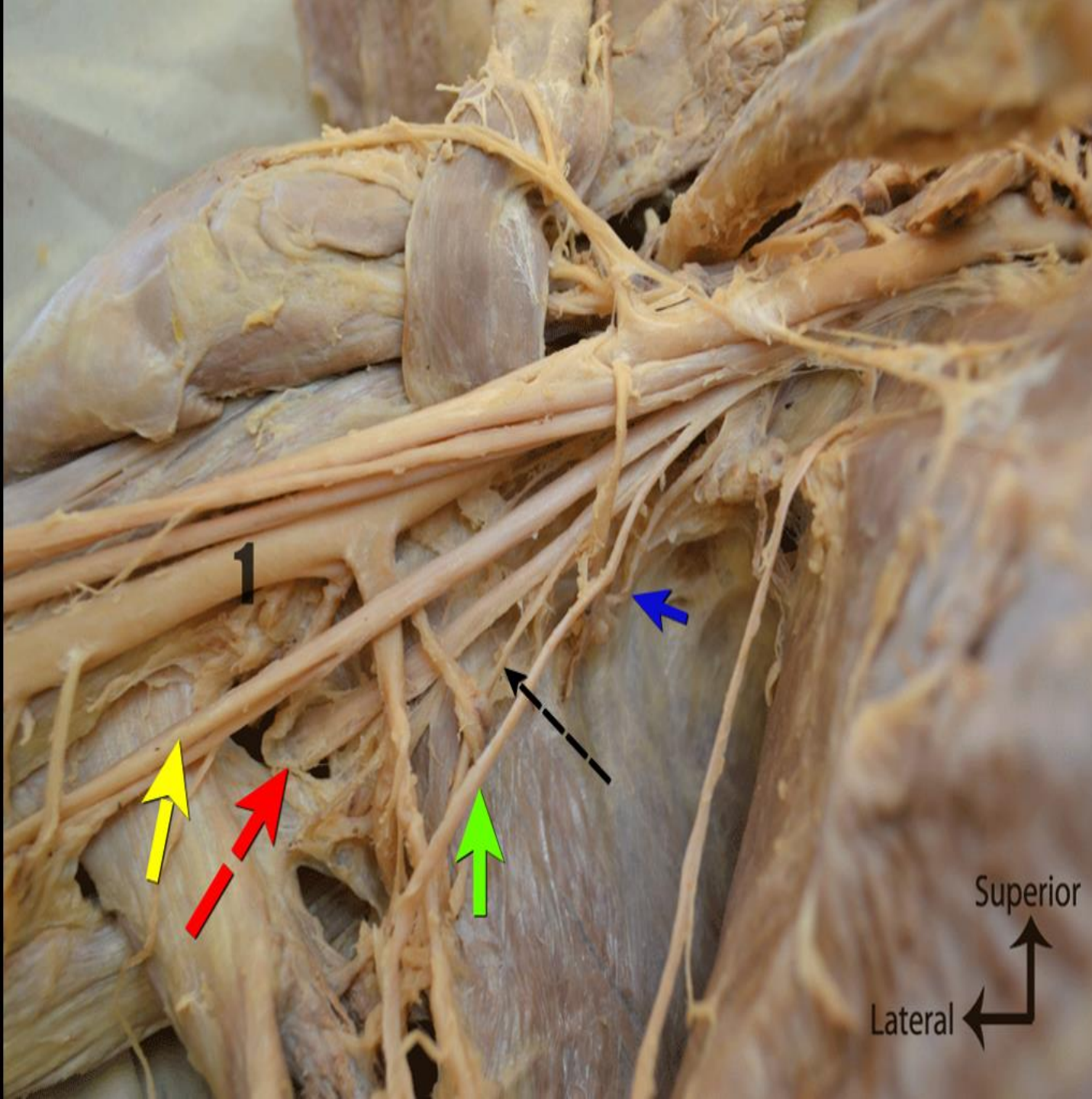
☐ **NERVES**







- 1= P. Major
- 2= P. minor
- 3= Coracobrachialis
- 4= Biceps(short head)
- 5= Biceps( long head)
- 6= Serratus ant.
- 7= Subscapularis
- 8= Teres major
- 9= Lat. Dorsi
- A= Musculocutaneous N
- B= Median N
- C=Ulnar N
- D= Long thoracic N



1= Axillary artery

Yellow arrow= Radial nerve

(gives muscular br. For long head of triceps while it in the axilla)

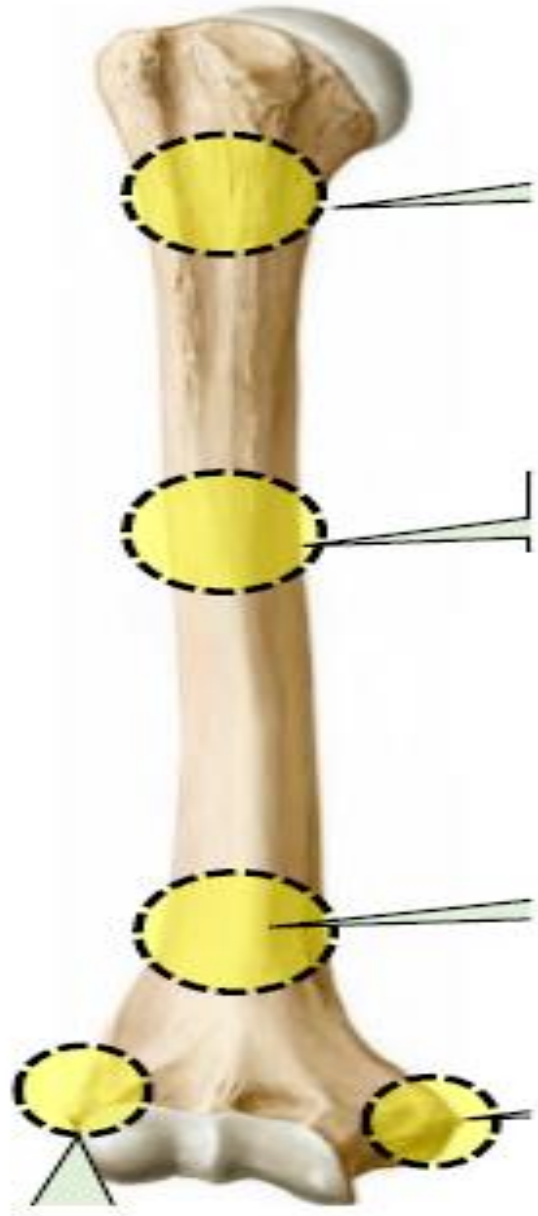
red arrow = axillary nerve (entering quadrangular triangle)

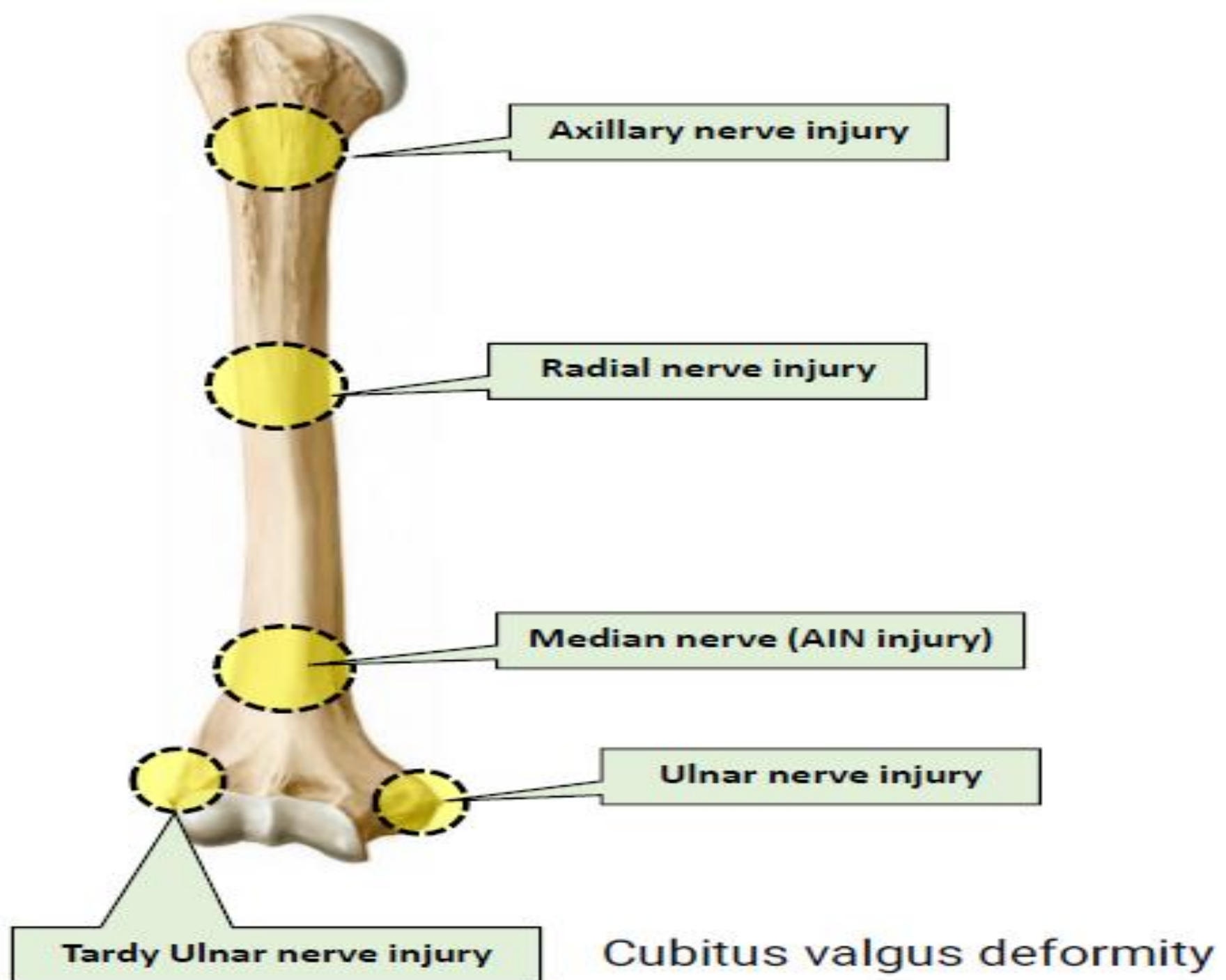
Green arrow= thoracodorsal nerve

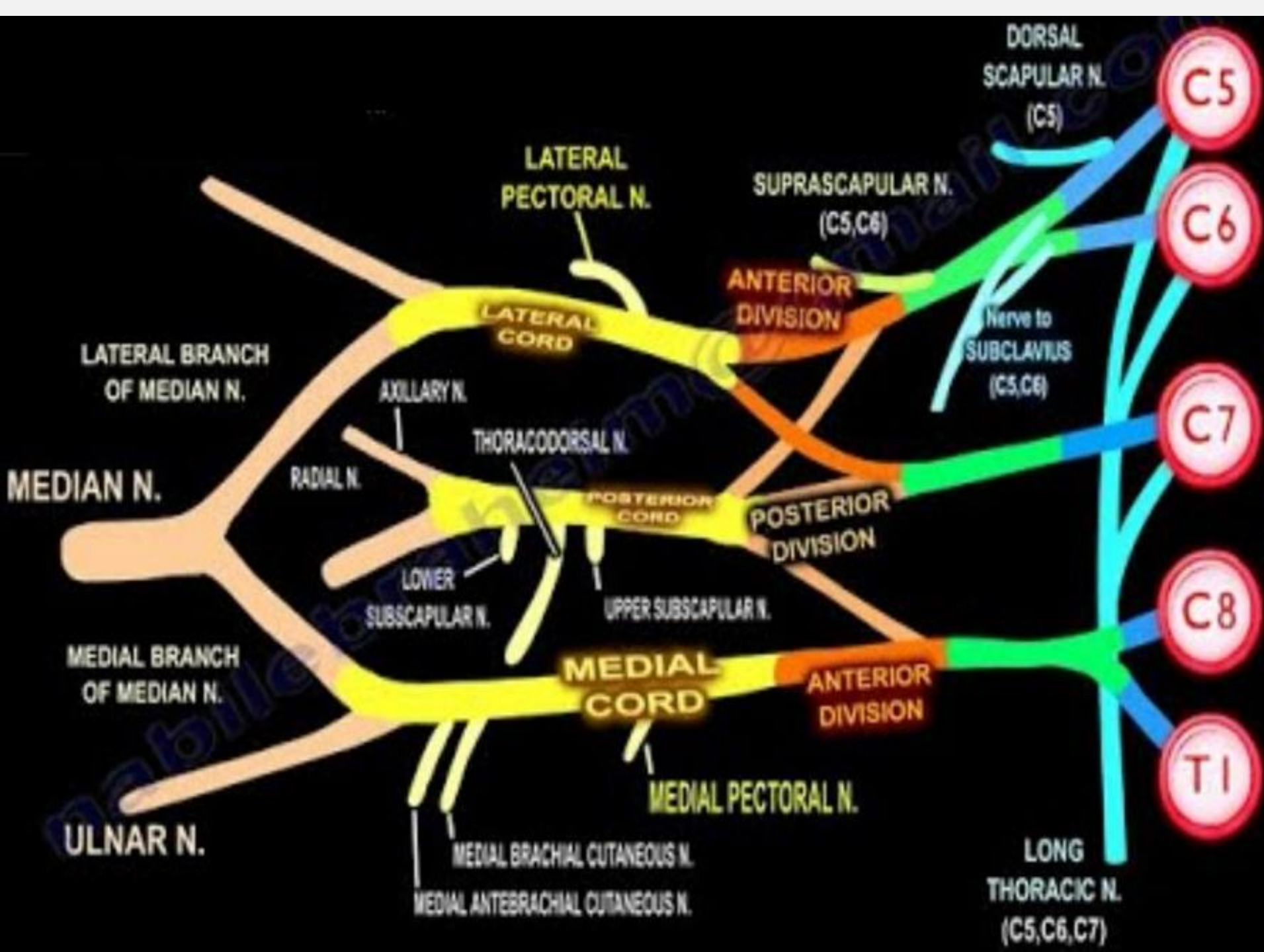
Black arrow= lower subscapular nerve

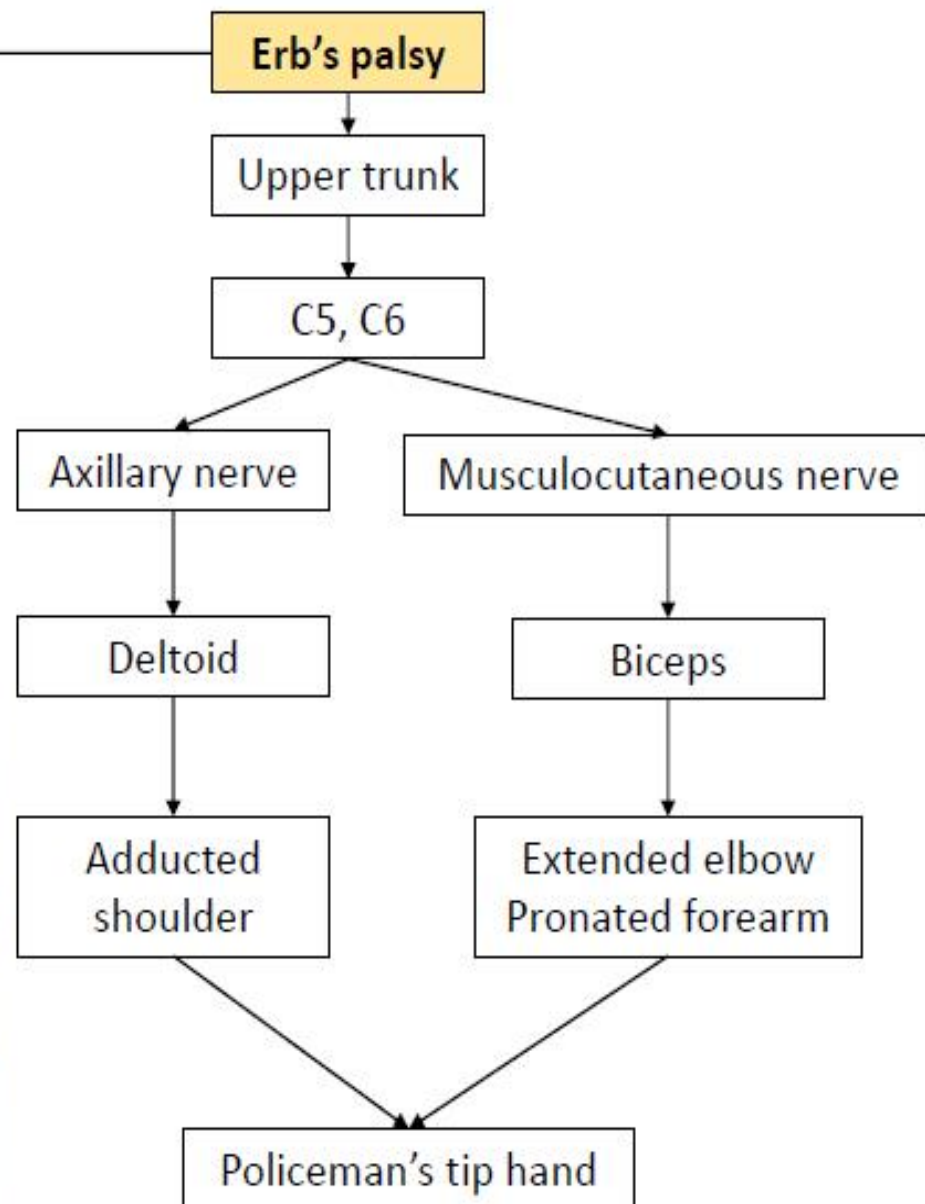
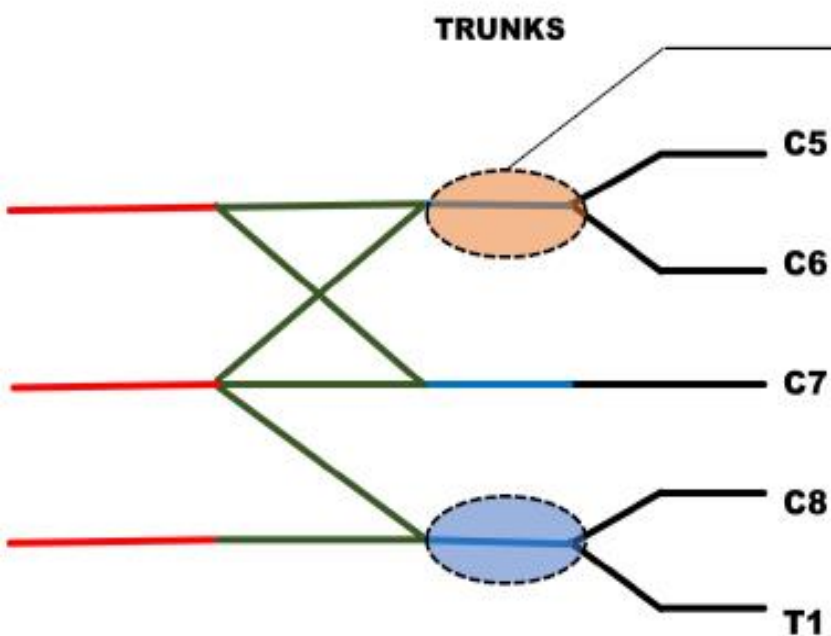
Blue arrow= upper subscapular nerve

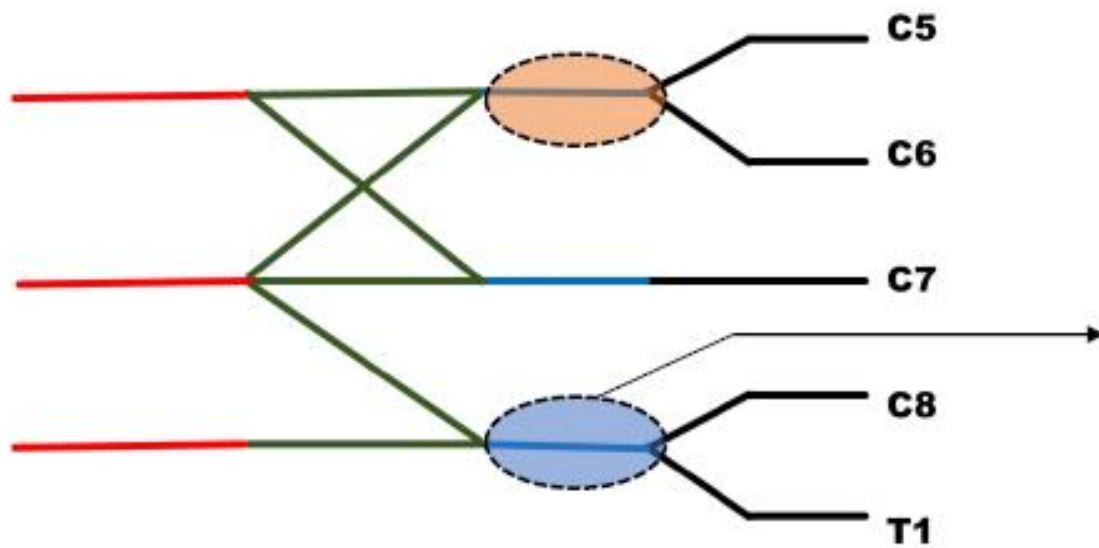












**Klumpke's palsy**

Lower trunk

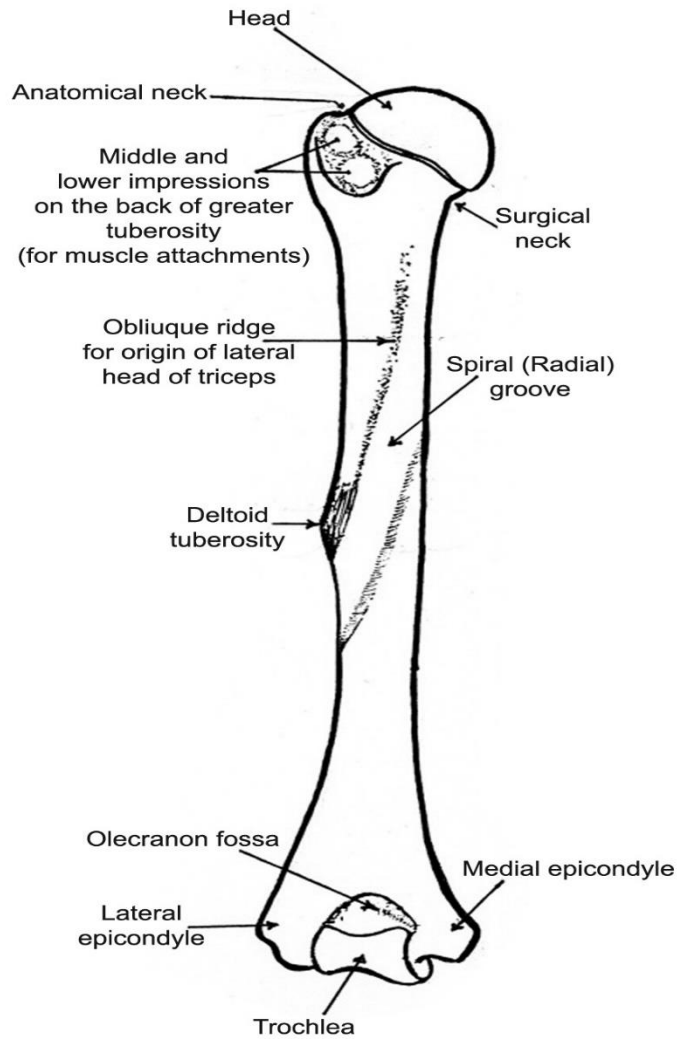
C8, T1

Intrinsic muscles of hand

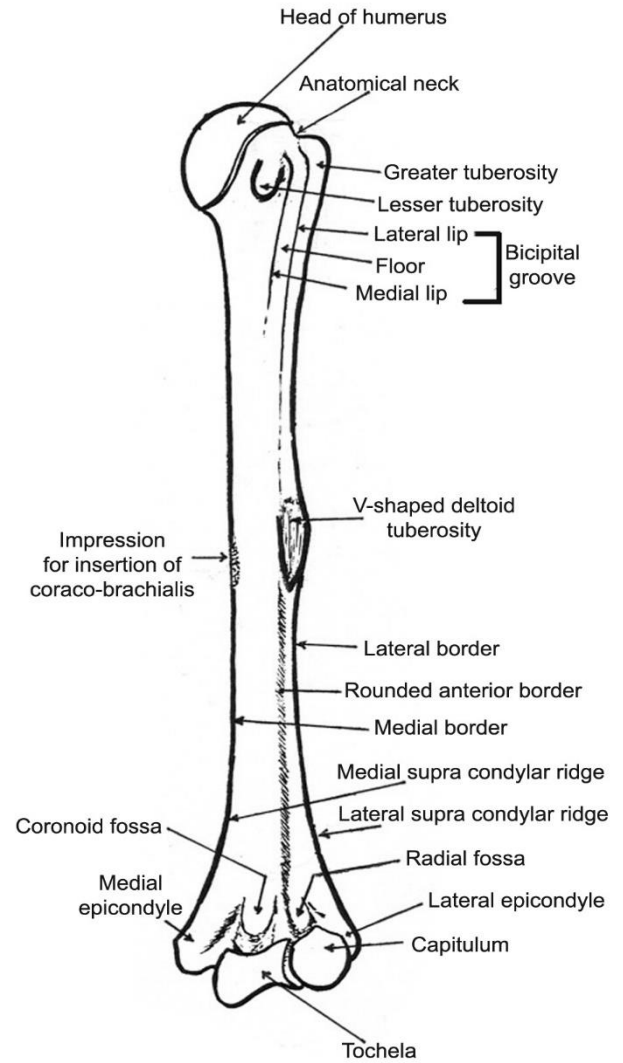
Hyperextension at MCP joints  
Flexion at IP joints

**CLAW HAND**

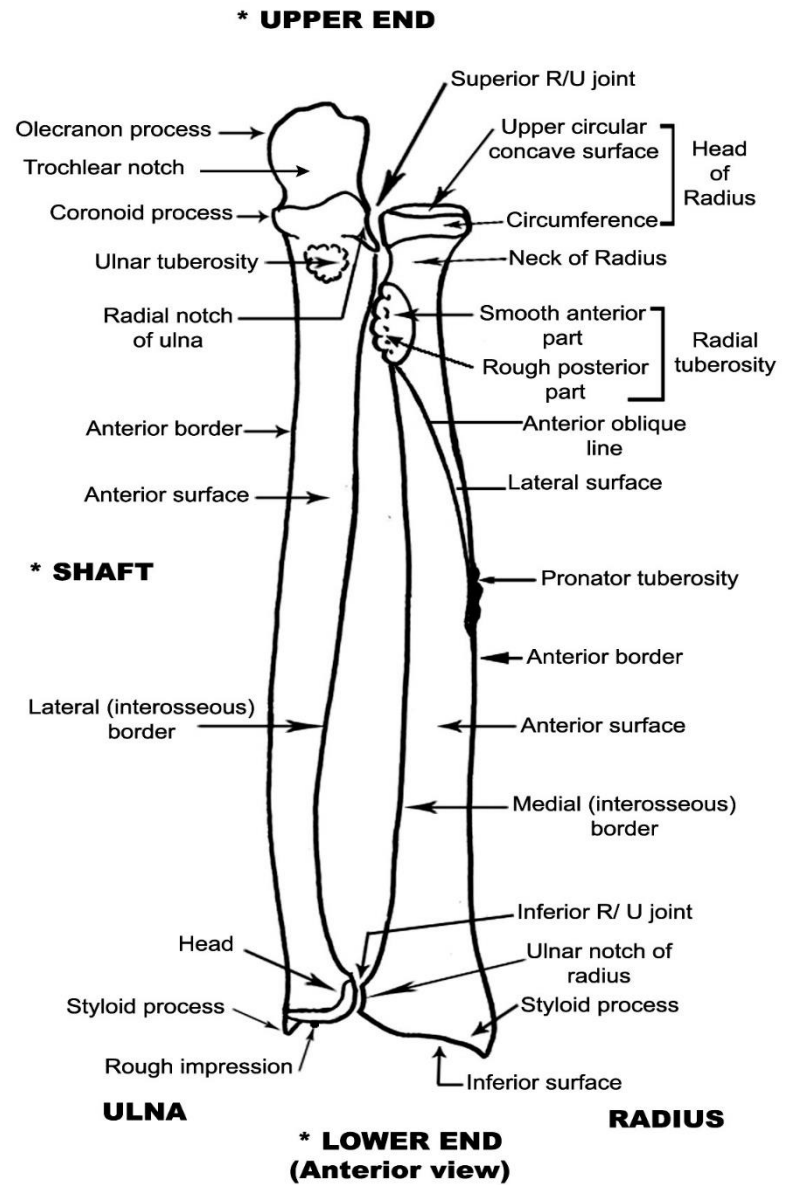
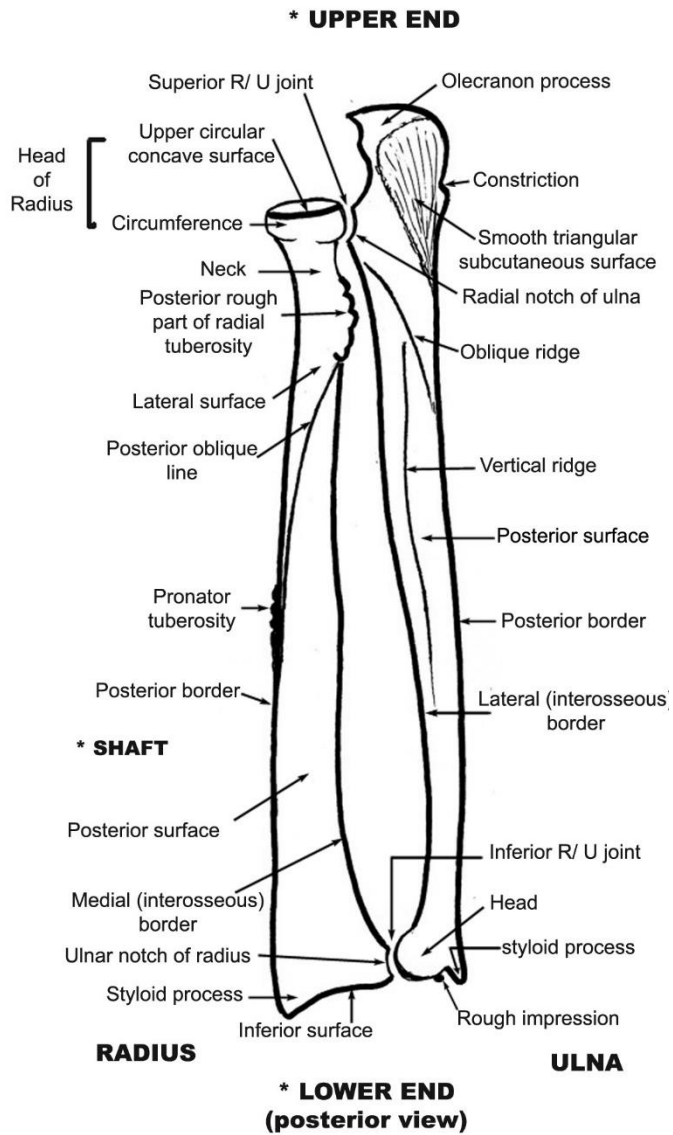




**HUMERUS**  
**(Posterior view)**



**HUMERUS**  
**(Anterior view)**



Olecranon

Radial notch of ulna

Head of radius

Neck of radius

Radial tuberosity

Trochlear notch

Coronoid process

Ulnar tuberosity

**Ulna**

**Radius**

Interosseous borders

Interosseous membrane

Ulnar notch of radius

Head of ulna

Styloid process

Articular facets

Styloid process

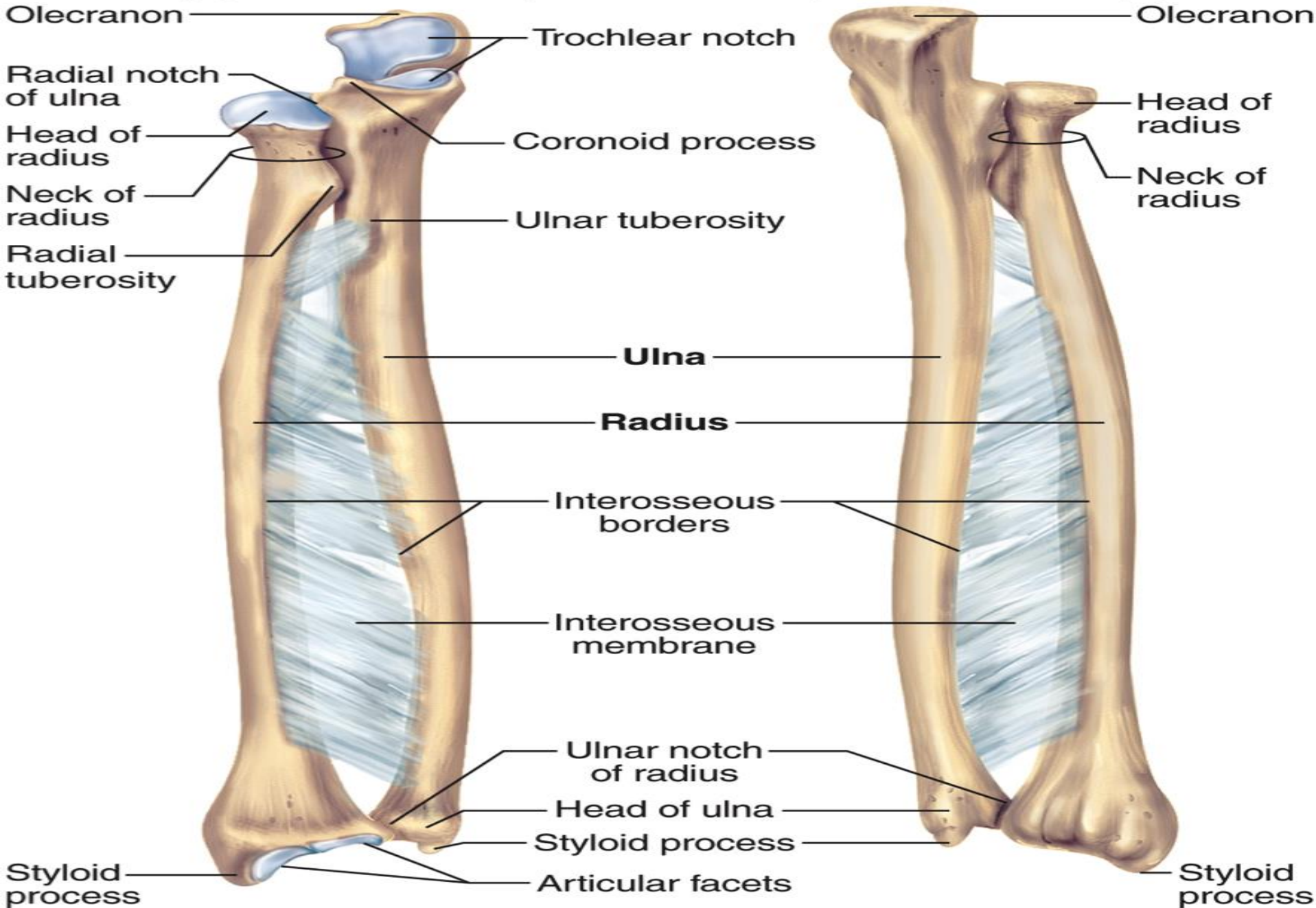
Olecranon

Head of radius

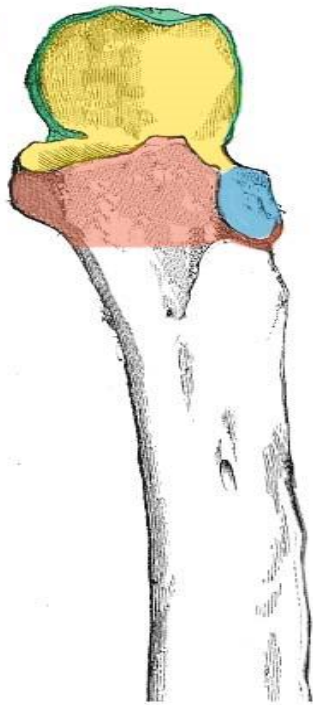
Neck of radius

**(a) Anterior view**

**(b) Posterior view**



**Anterior:**

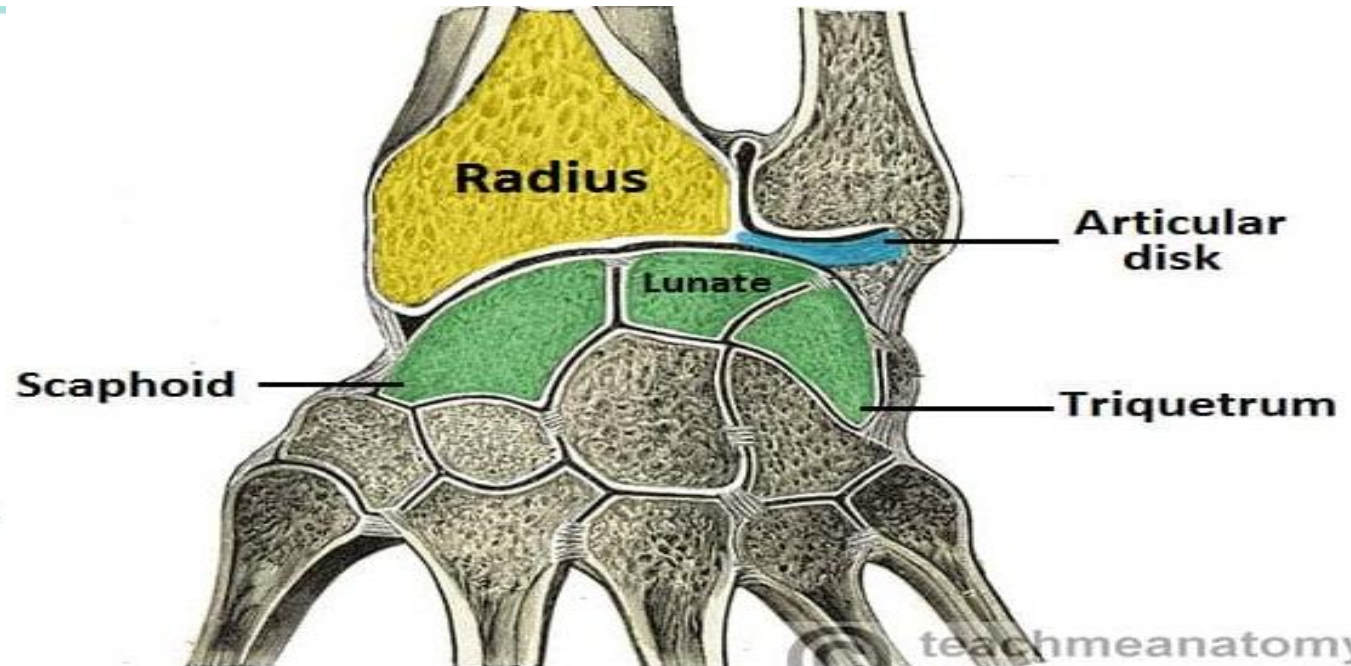
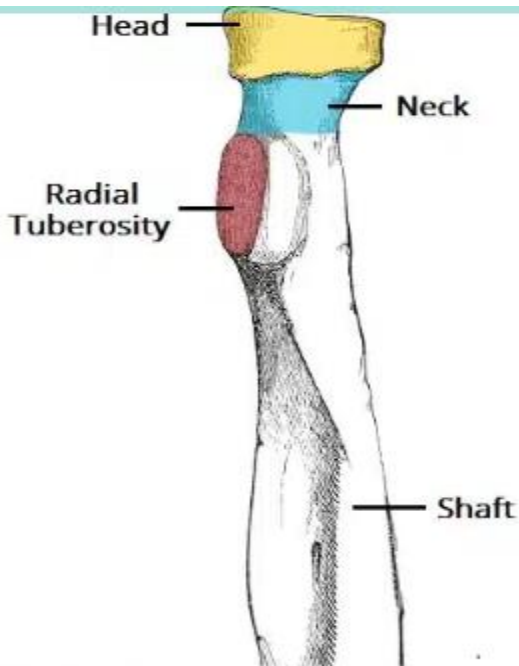


**Lateral:**

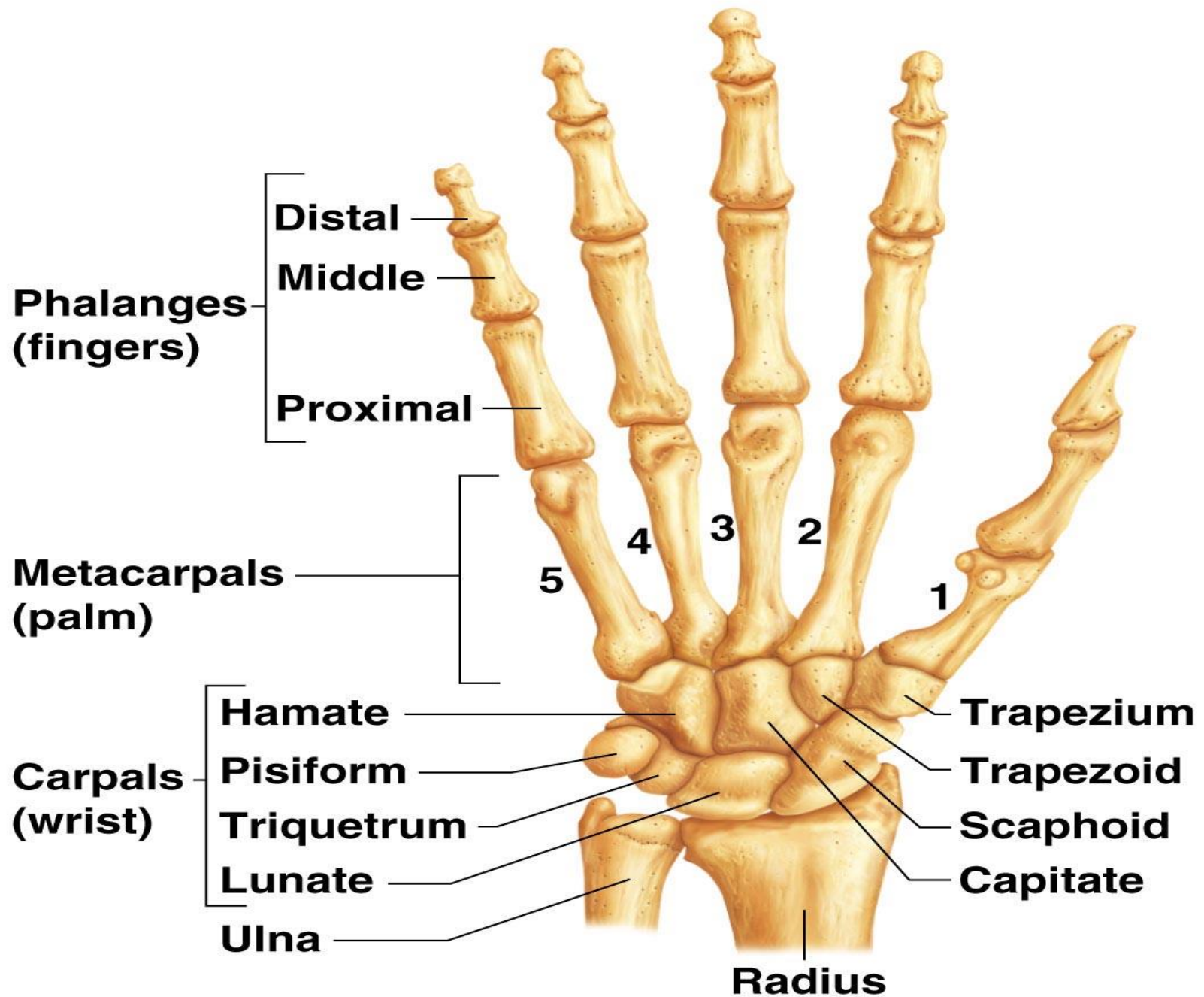


-  Olecranon
-  Coronoid Process
-  Trochlear Notch
-  Radial Notch

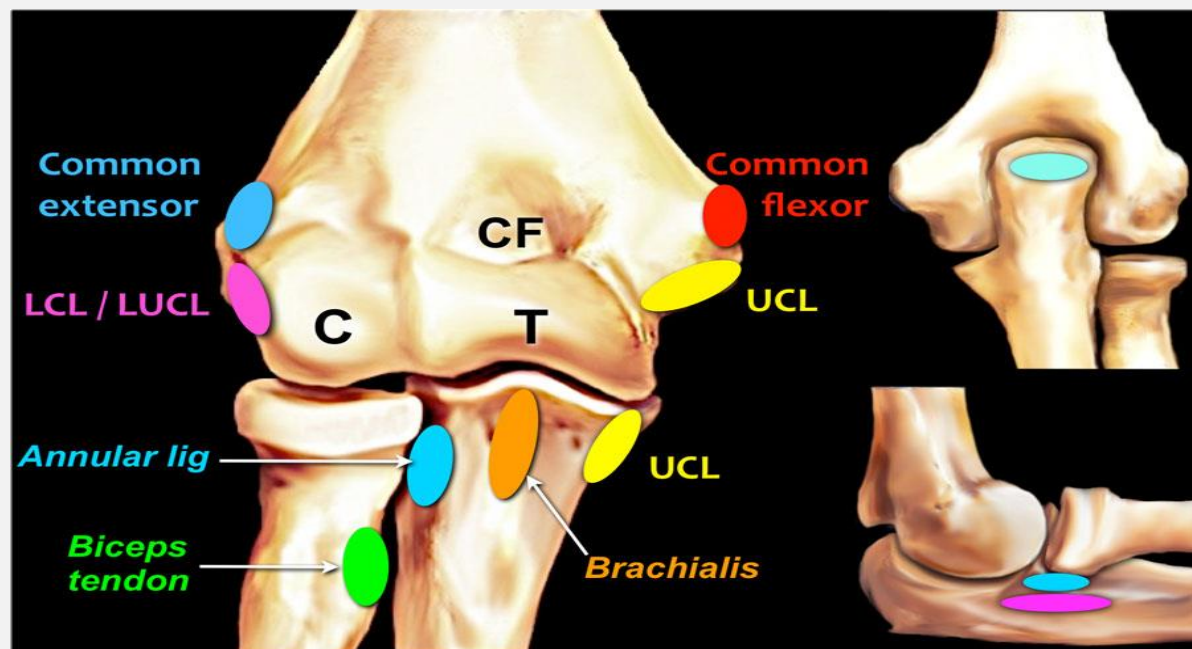
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The #1 Applied Human Anatomy Site on the Web.



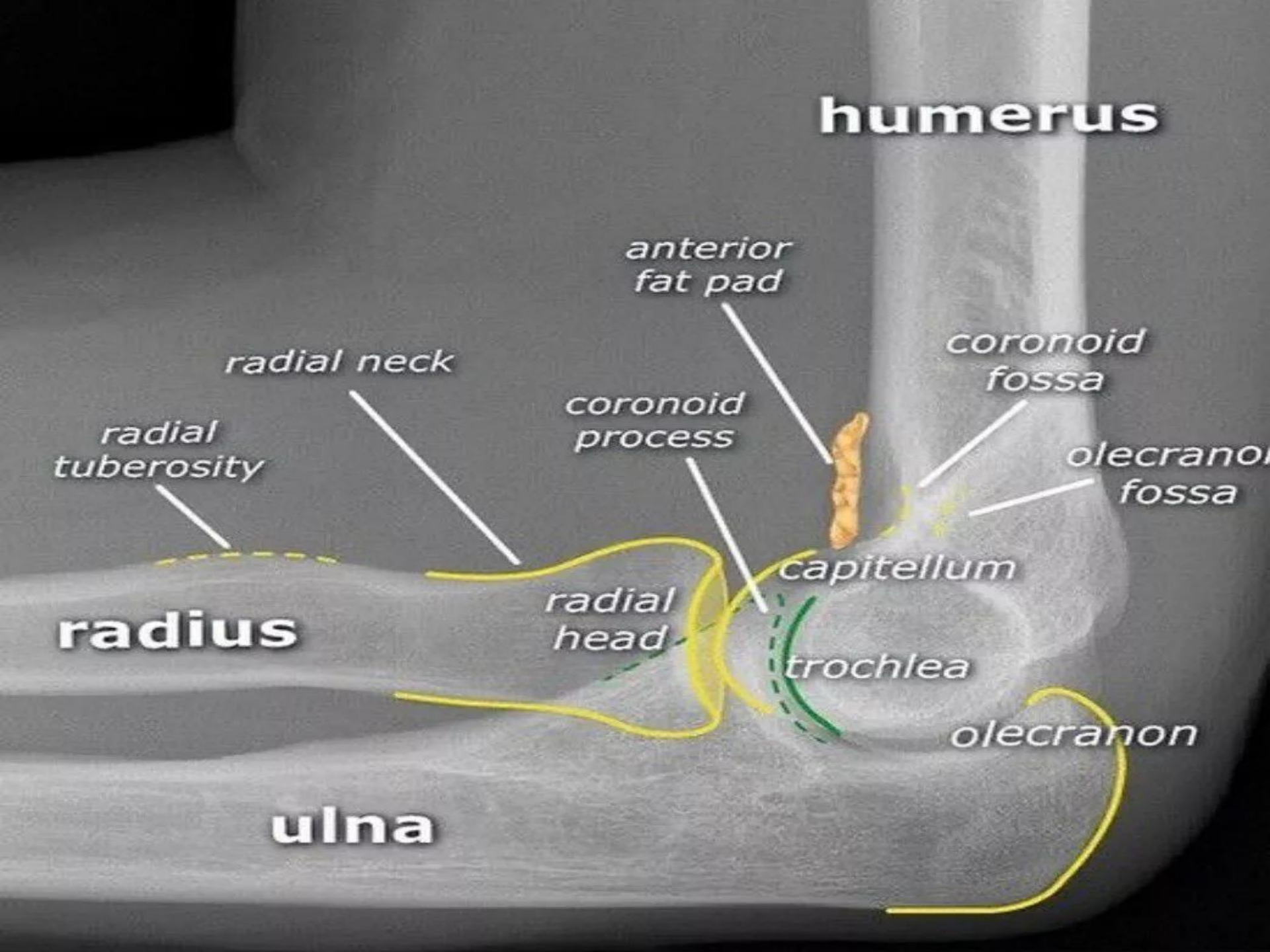
teachmeanatomy







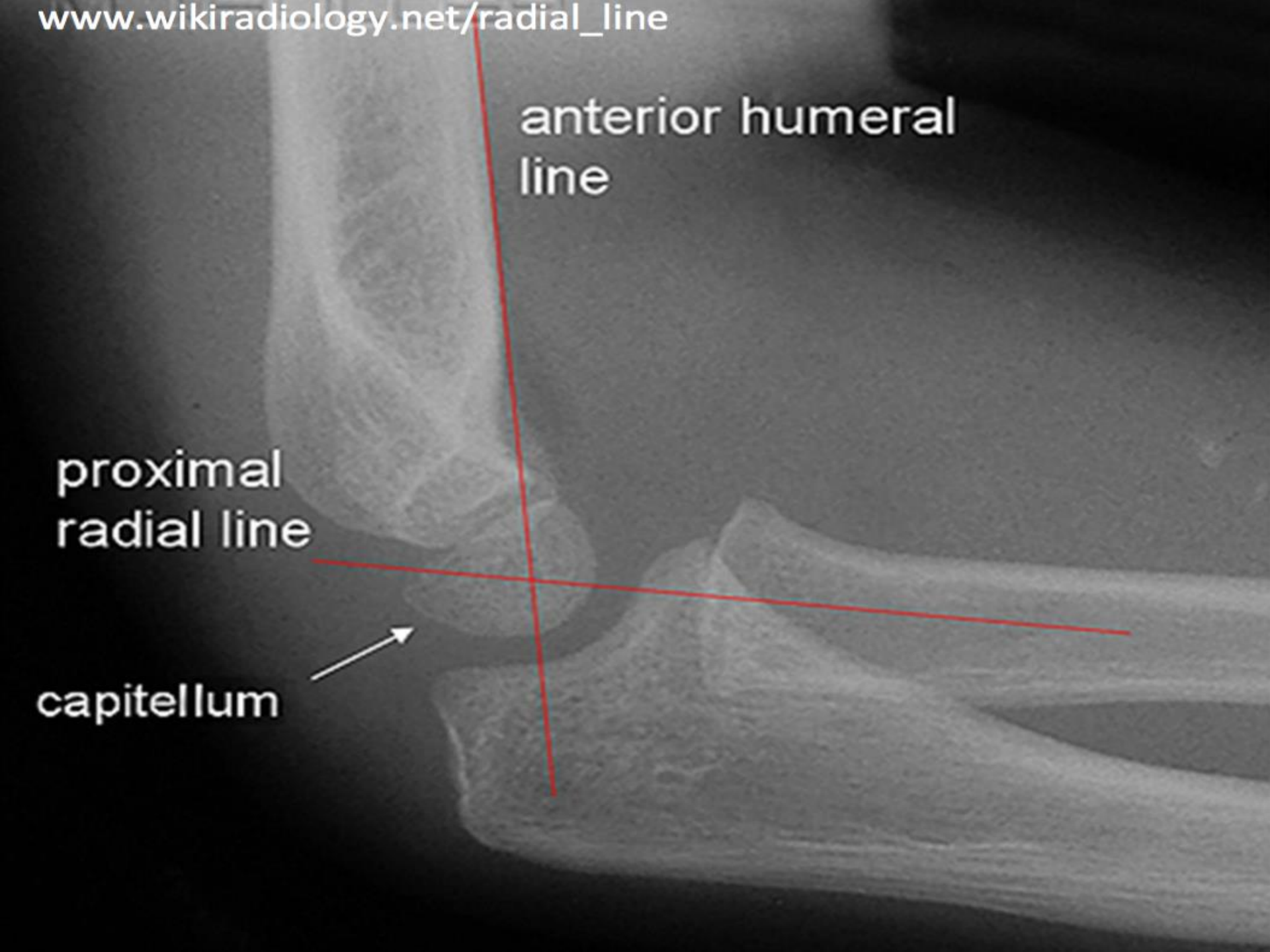




anterior humeral  
line

proximal  
radial line

capitellum





\*



NK

\*



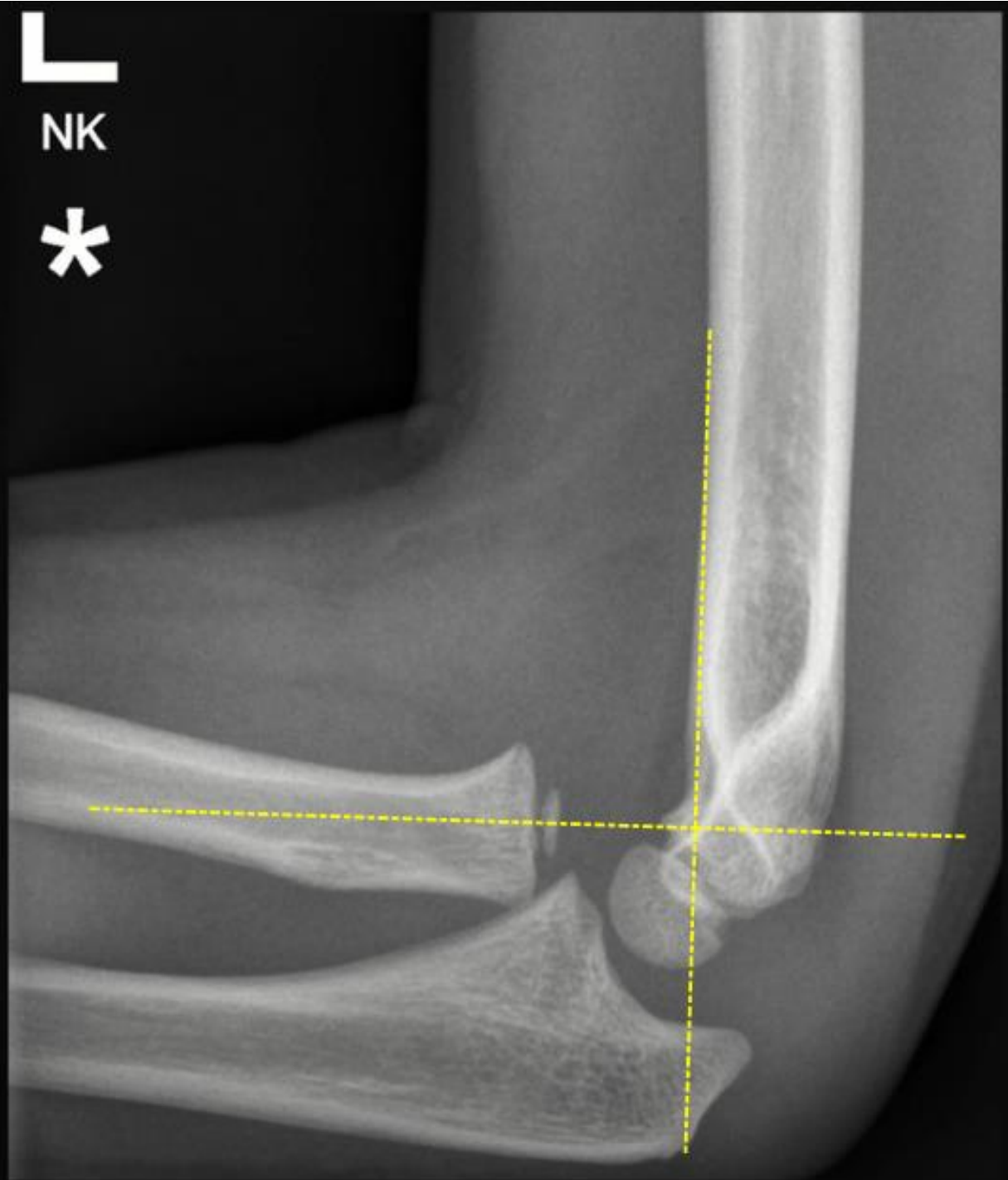


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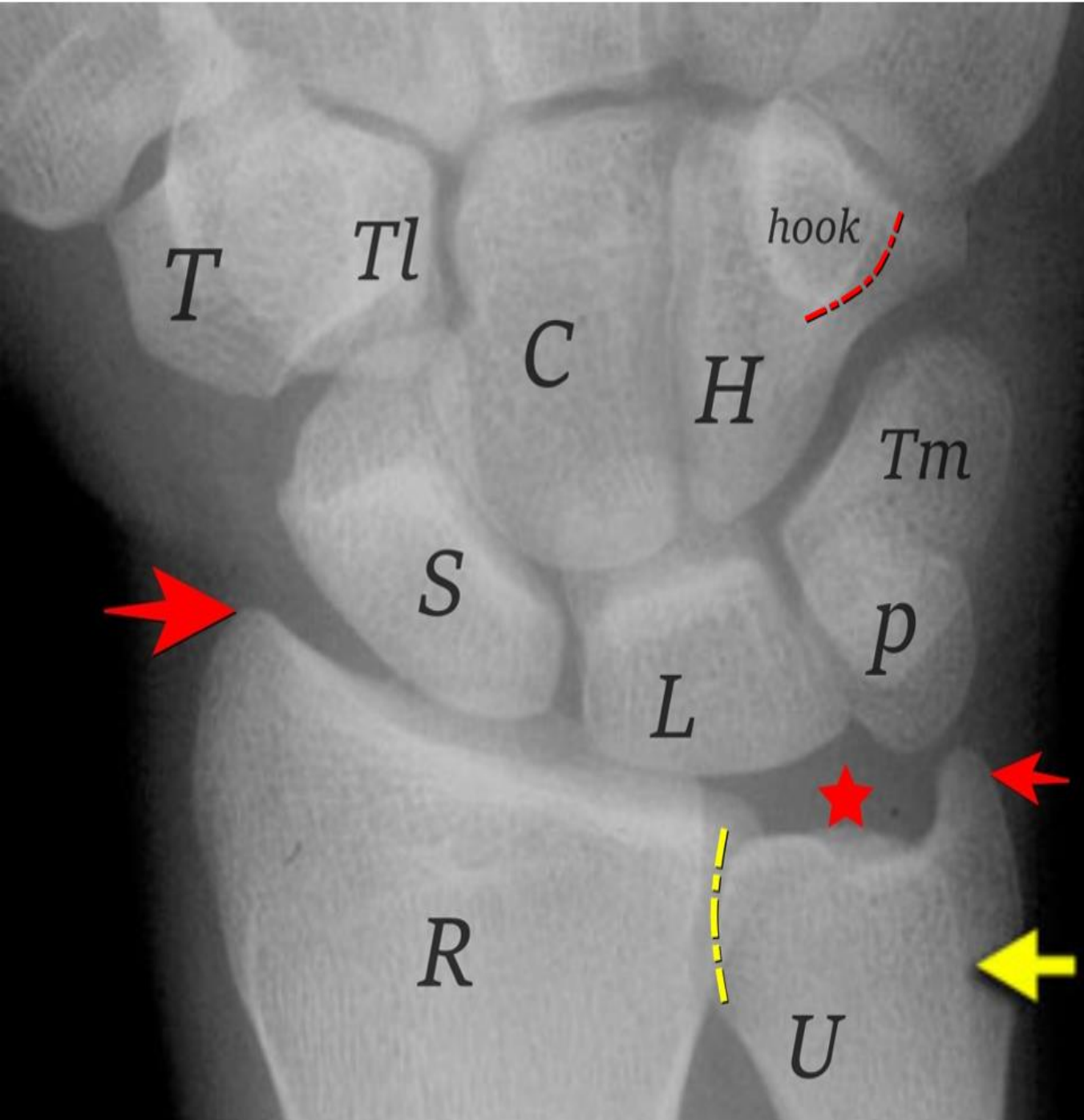


NK

\*







### AP x-ray , wrist joint

- **Red arrows**= styloid processes
- **Red asterisk**= triangular fibrocartilage complex(**TFCC**)
- **Dotted line**= distal radioulnar joint
- **Yellow arrow**= ECU groove seen radial to the midportion of the ulnar styloid
- **Red dotted arrow**= groove for deep branch of ulnar nerve

# **MUSCLES OF FOREARM**

**NOTE:** There are 20 mm. in the forearm, to remember them, follow the **8** rules

According to their actions, the muscles of flexor compartment can classify into **FOUR** groups

## RULE 1



### WRIST FLEXION

Flexor carpi ulnaris  
Flexor carpi radialis



### THUMB FLEXION

Flexor pollicis longus



### FINGER FLEXION

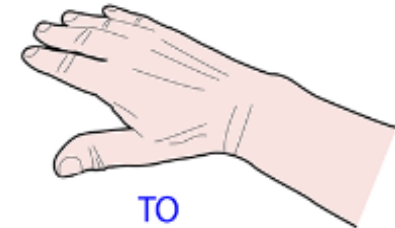
Flexor digitorum superficialis  
Flexor digitorum profundus

**Note:**

Palmaris longus excluded



FROM  
SUPINATION

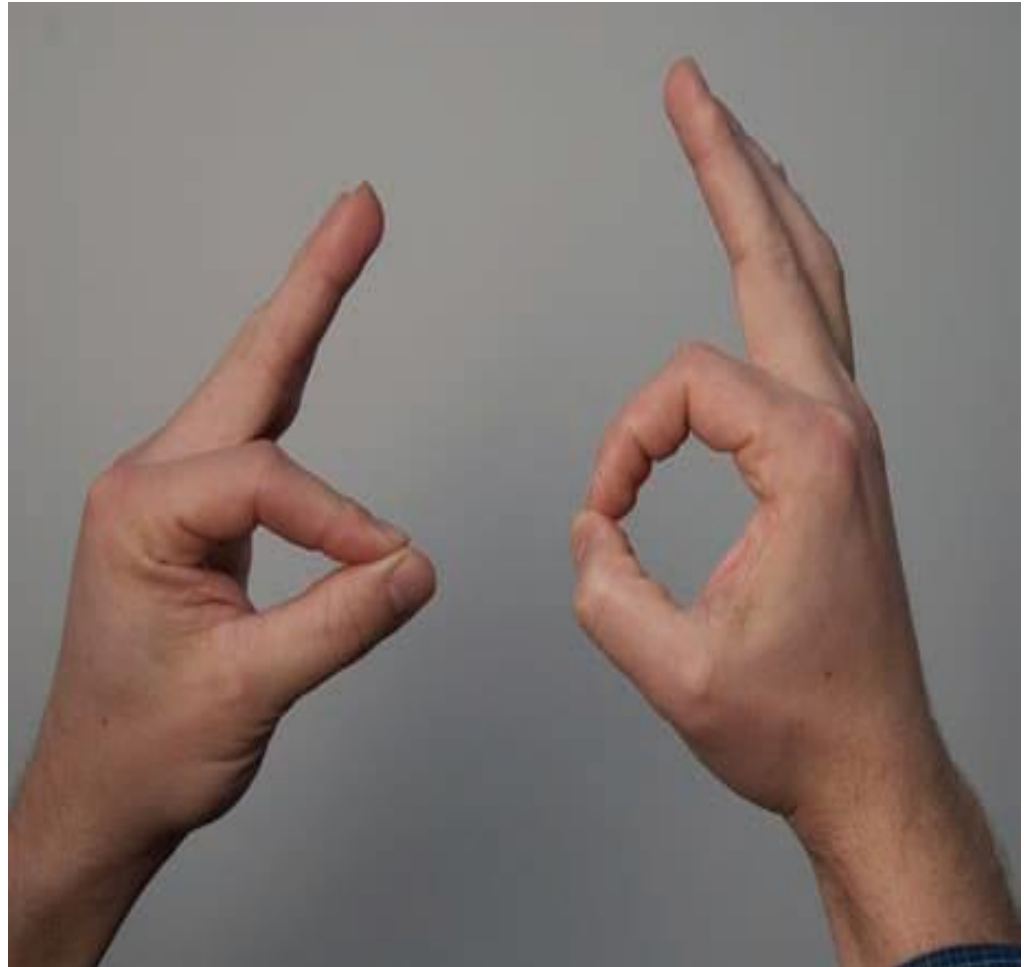


TO  
PRONATION

Pronator teres  
Pronator quadratus

## RULE 2

- All muscles of flexor compartment supplied directly by **MEDIAN** nerve ***or*** indirectly through its branch **anterior interosseous (AIN)** **EXCEPT FCU & ½ FDP**-ulnar nerve
- **AIN** supply 3 mm.
  - 1)FPL
  - 2)Lateral ½ of FDP
  - 3)PQ
- **okay sign** is specific for examination integrity of AIN



Okay sign

## **RULE 3** Muscles of the posterior compartment

- **12** MM, according to their attachment, they are:

**3** carpi-----ECU, ECRL,ECRB

**3** pollicis -----Abd PL, EPL,EPB

**3** digit-----ED, EDM,EI

**3** OTHERS----- Brachioradialis, Supinator& Anconeus

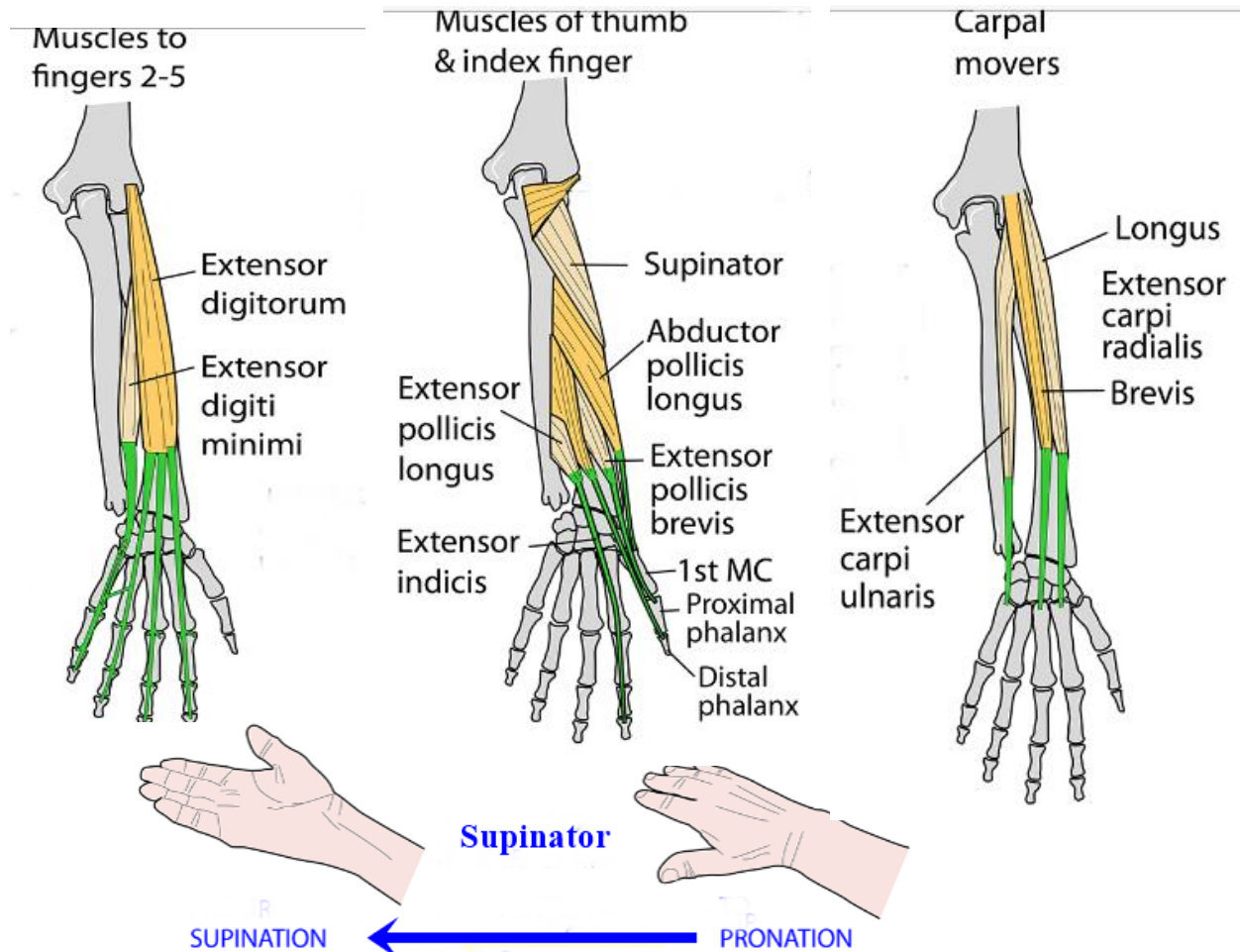
# RULE 4

According to their actions, the muscles of extensor compartment can classify into **FOUR** groups

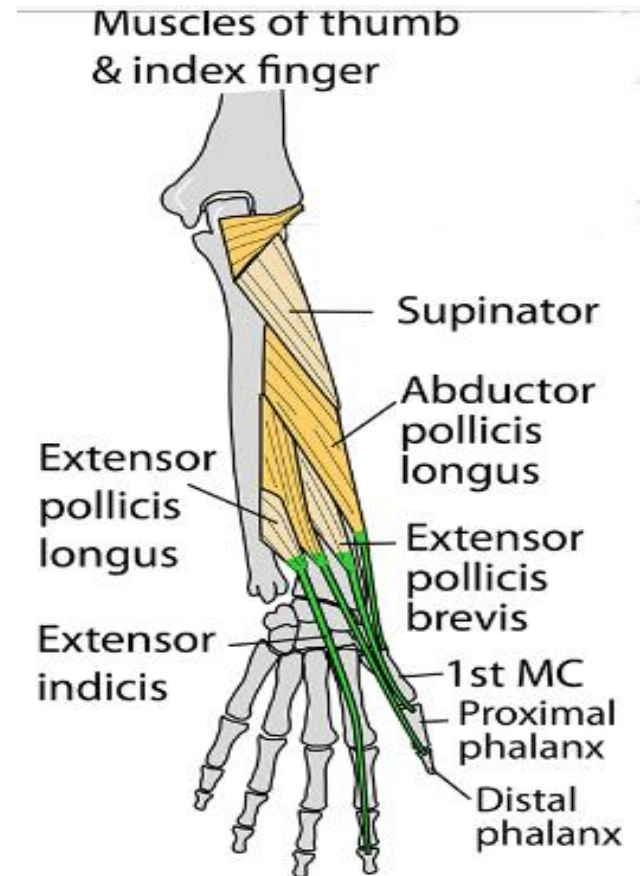
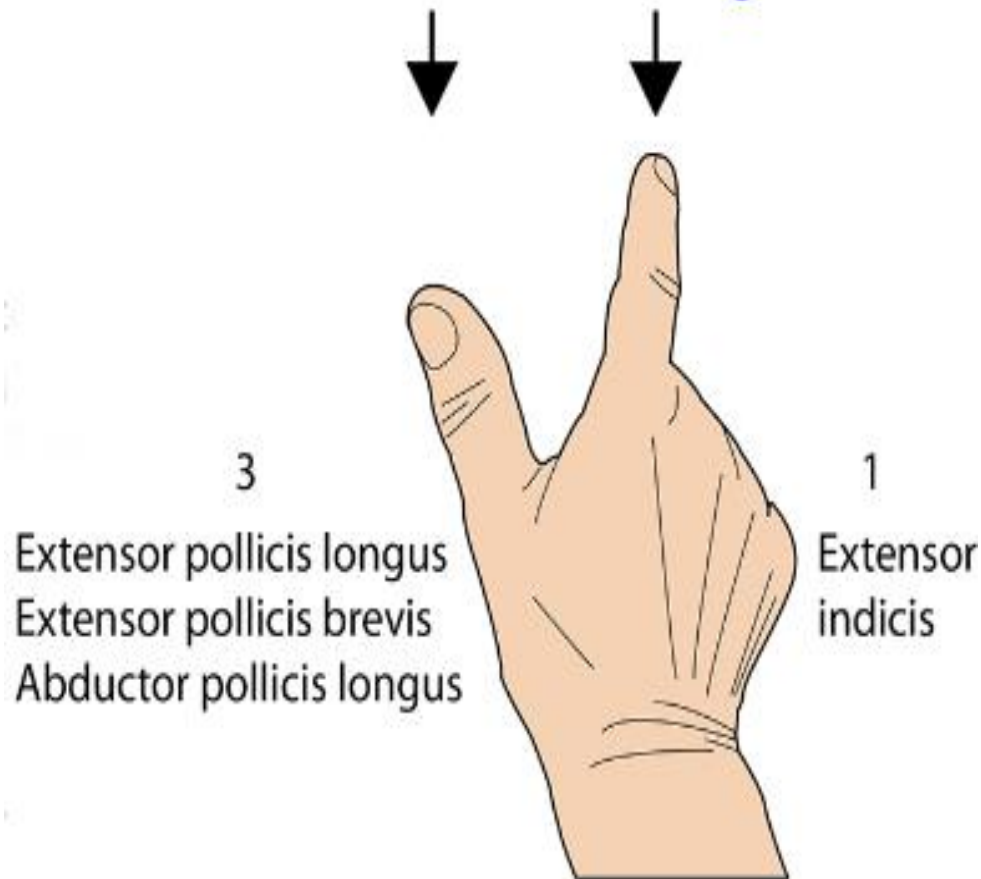
**Note:**

We excluded **TWO** other mm.

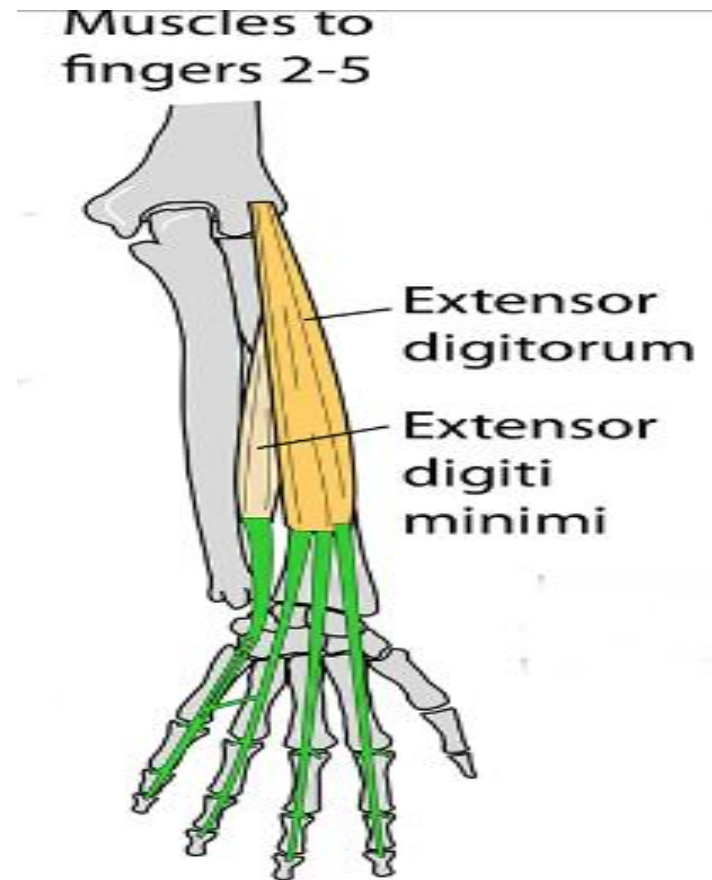
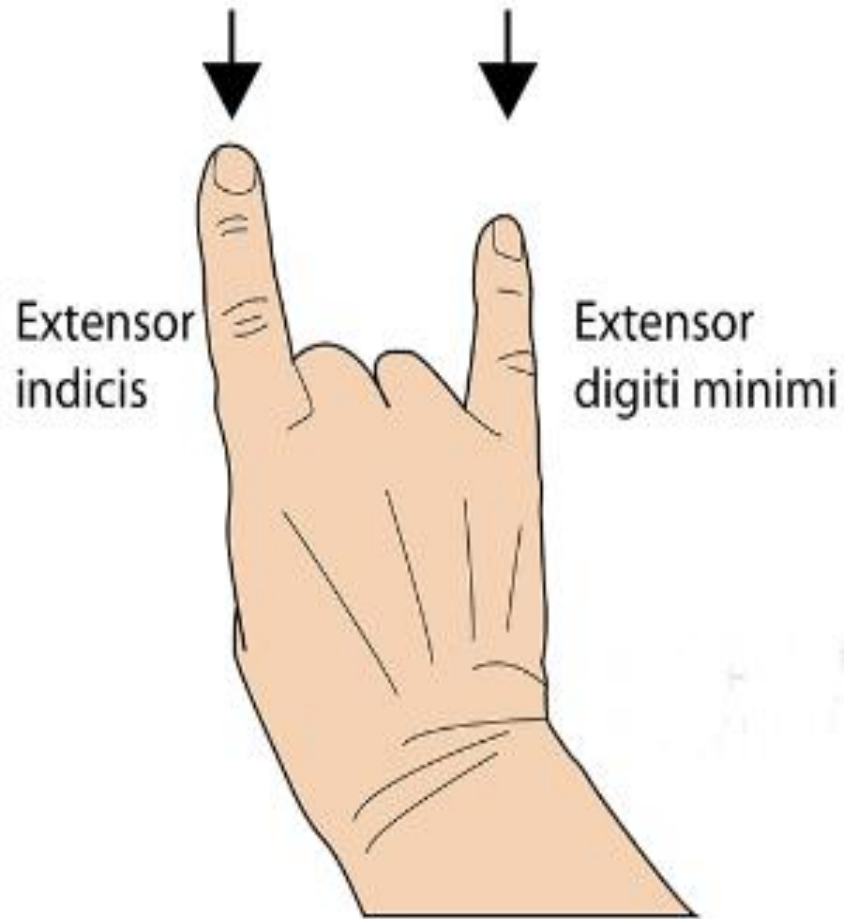
- Brachioradialis
- Anconeus



# RULE 5



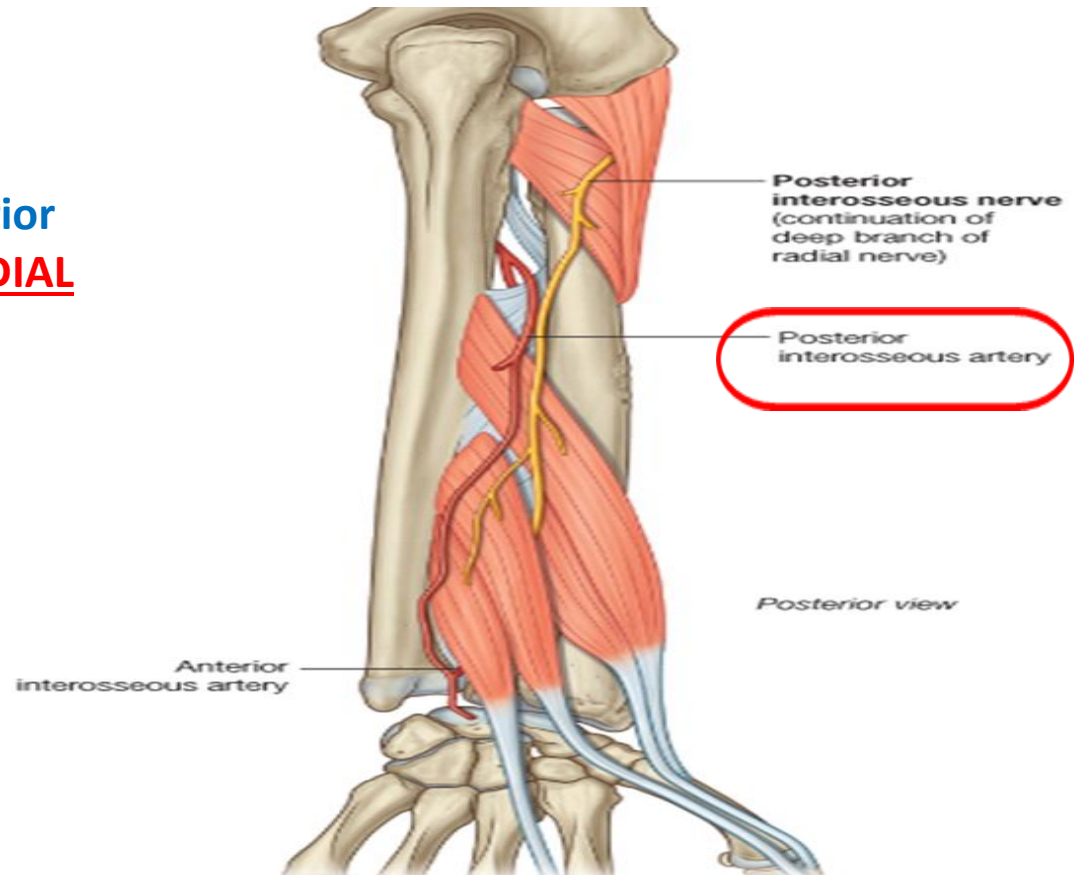
# RULE 6



# RULE 7

All muscles of extensor compartment supplied by **Posterior interosseous** which branch of **RADIAL nerve**

Radial nerve innervates the **BEST** directly( as it in spiral groove)  
**Brachioradialis**  
**Extensors**(ECRL)  
Anconus



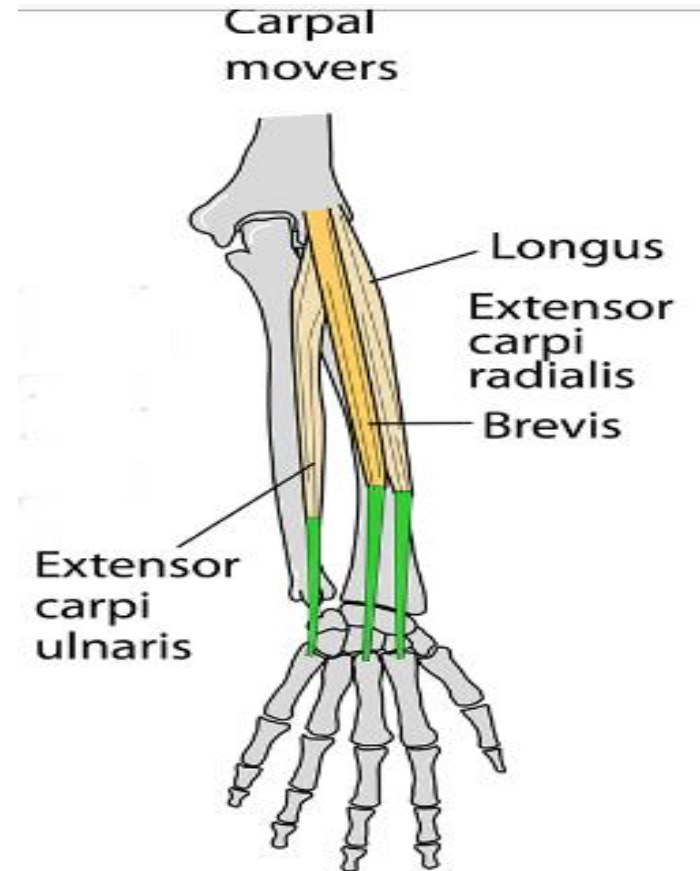
# RULE 8

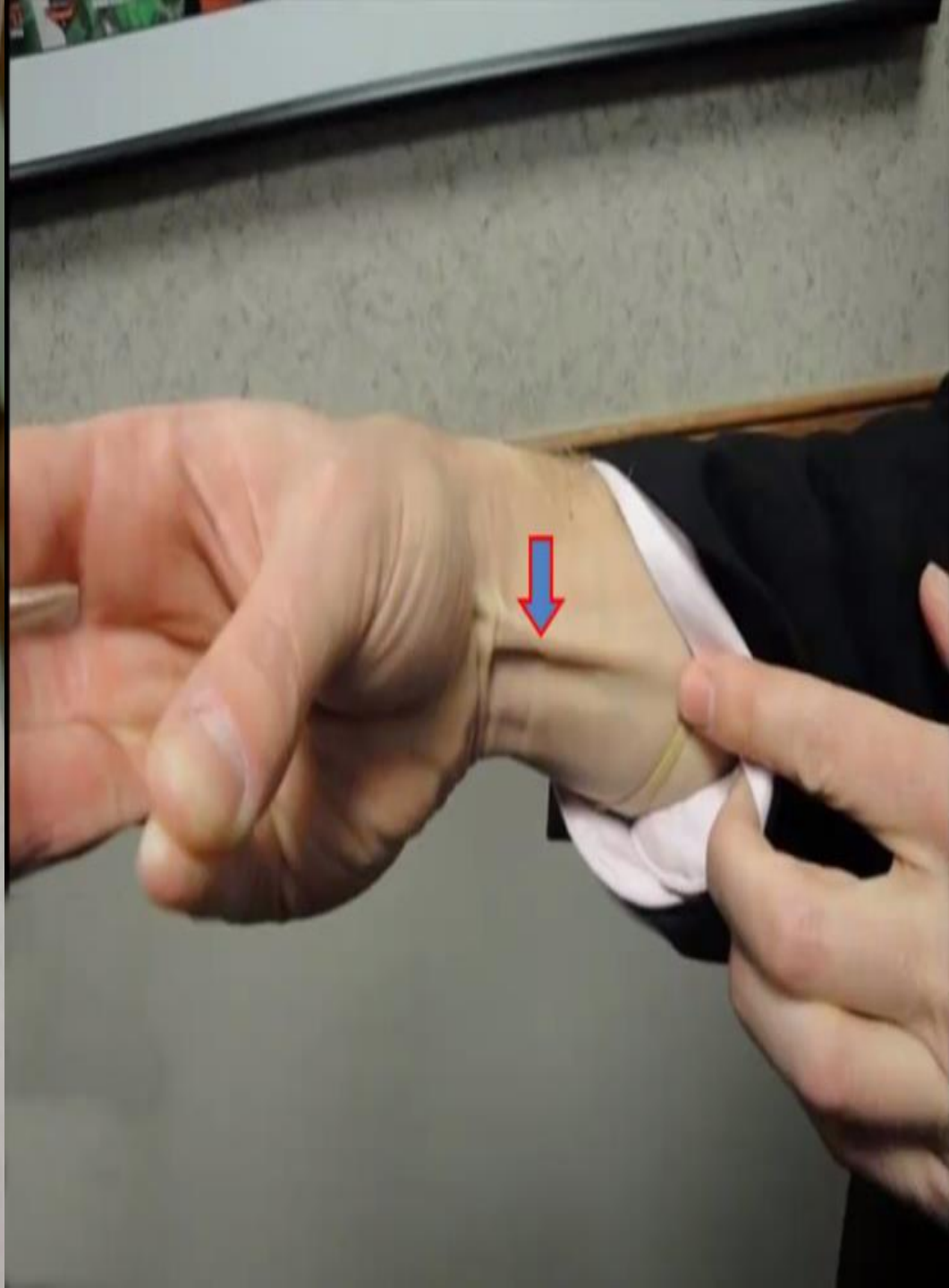
❑ ECU & ECR : their coresponders: FCU & FCR have same insertion.

❑ ECU&FCU are adductor to wrist also

❑ ECR&FCR are abductor to wrist also

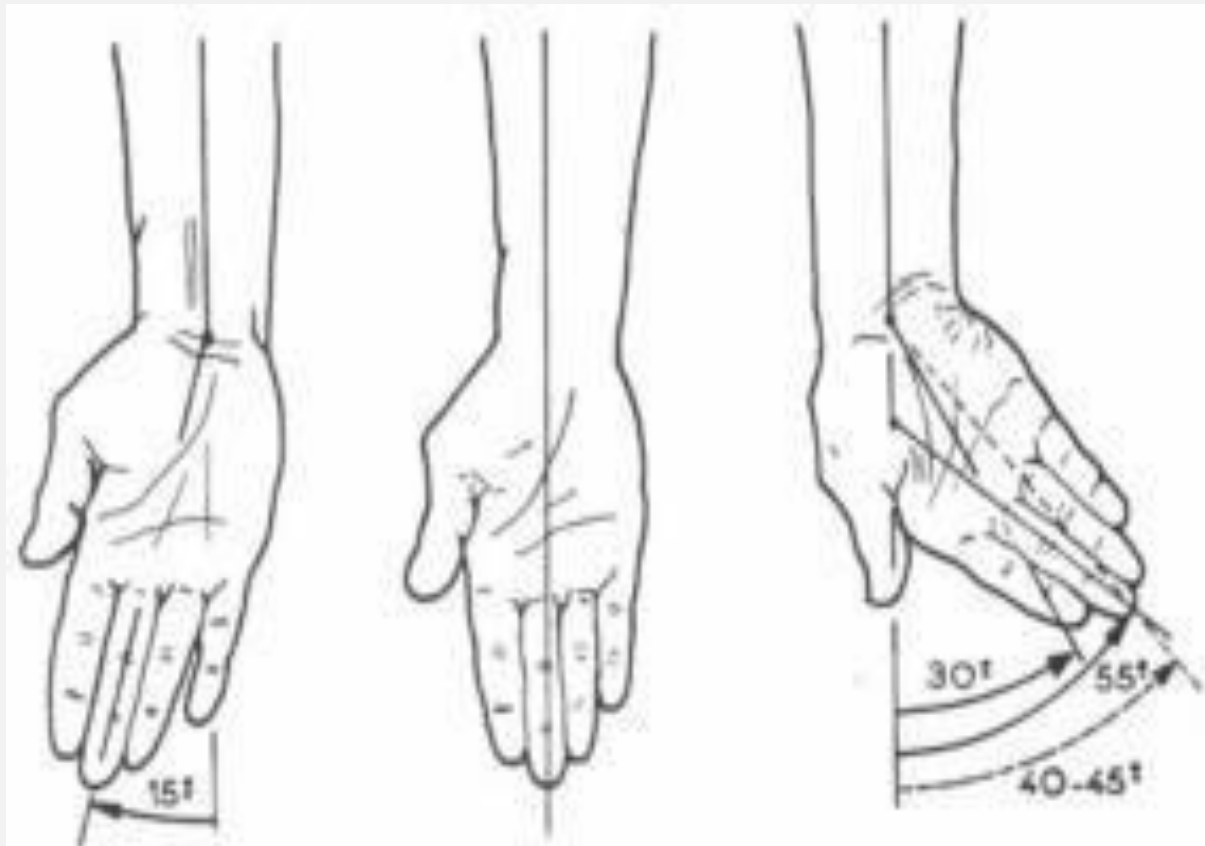
❑ ED attaches to the base of proximal phalanx while FDS& FDP do not (so ED extends MP joints only ,don't act on PIP&DIP)





## ❑ range of adduction at wrist joint more than in abduction?

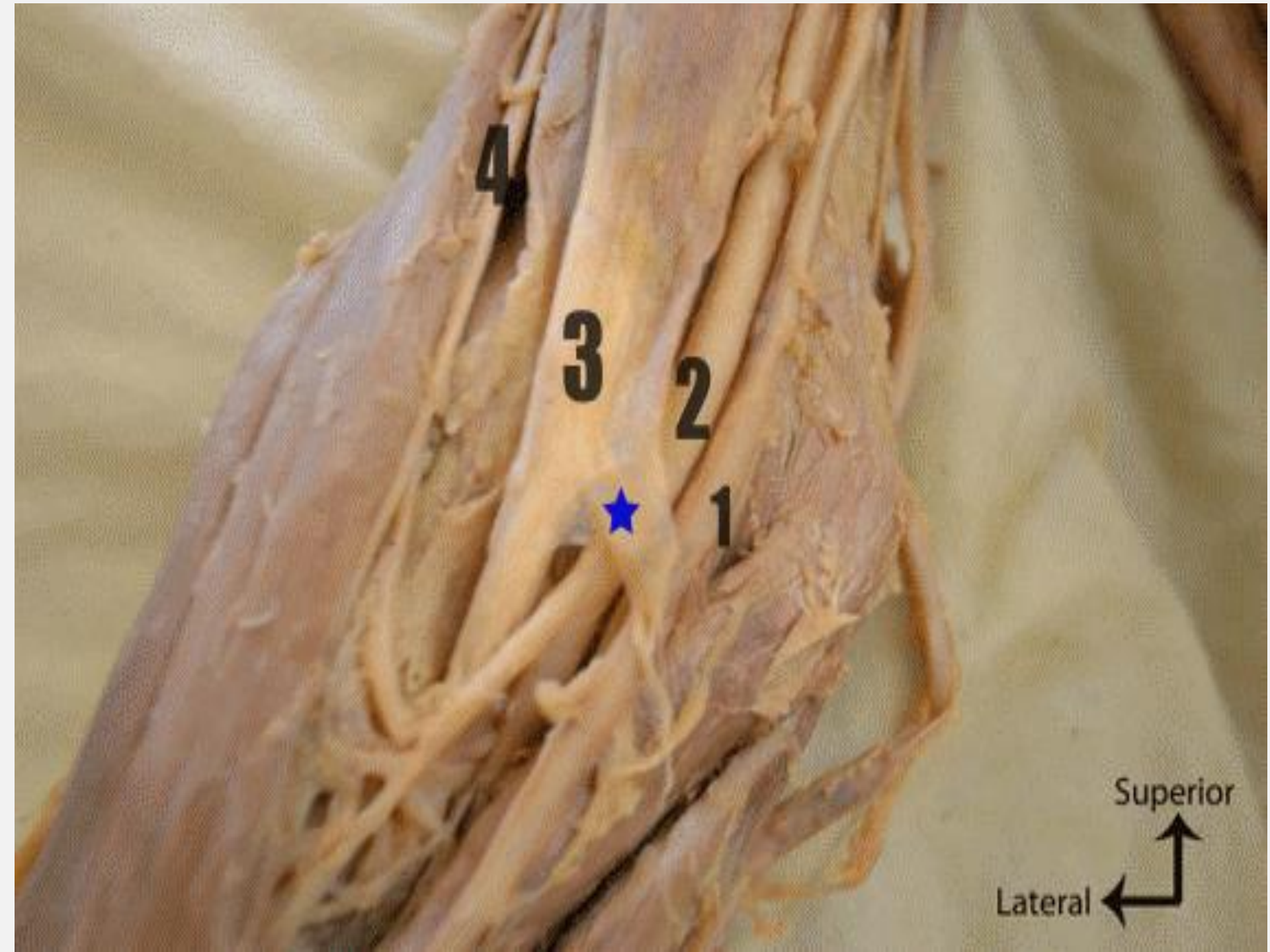
- ❑ Because the more distal site of radial styloid process restricts abduction
- ❑ Adduction – Produced by **ECU& FCU**
- ❑ Abduction – Produced by abductor pollicis longus, flexor carpi radialis, extensor carpi radialis longus and brevis.

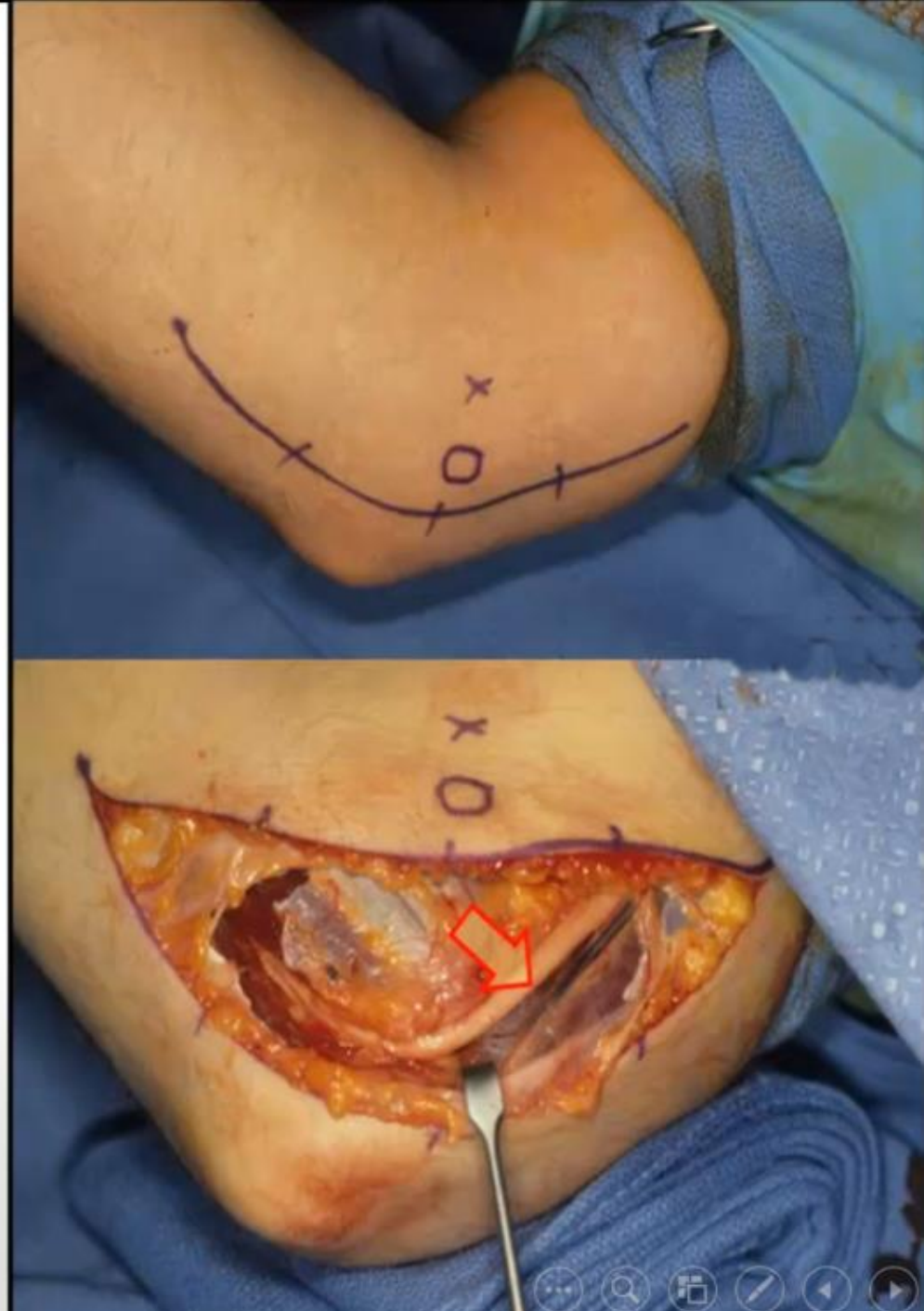
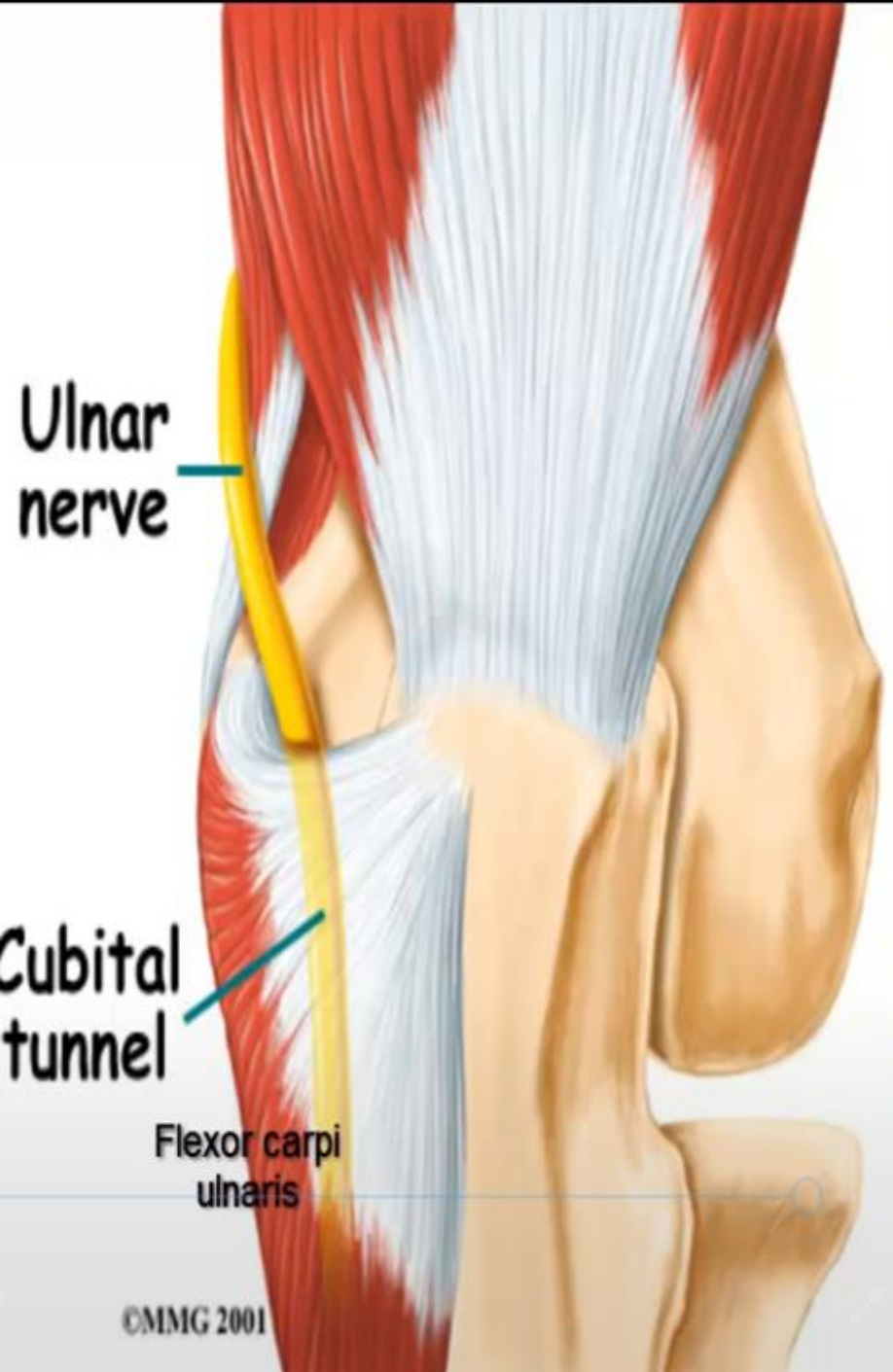


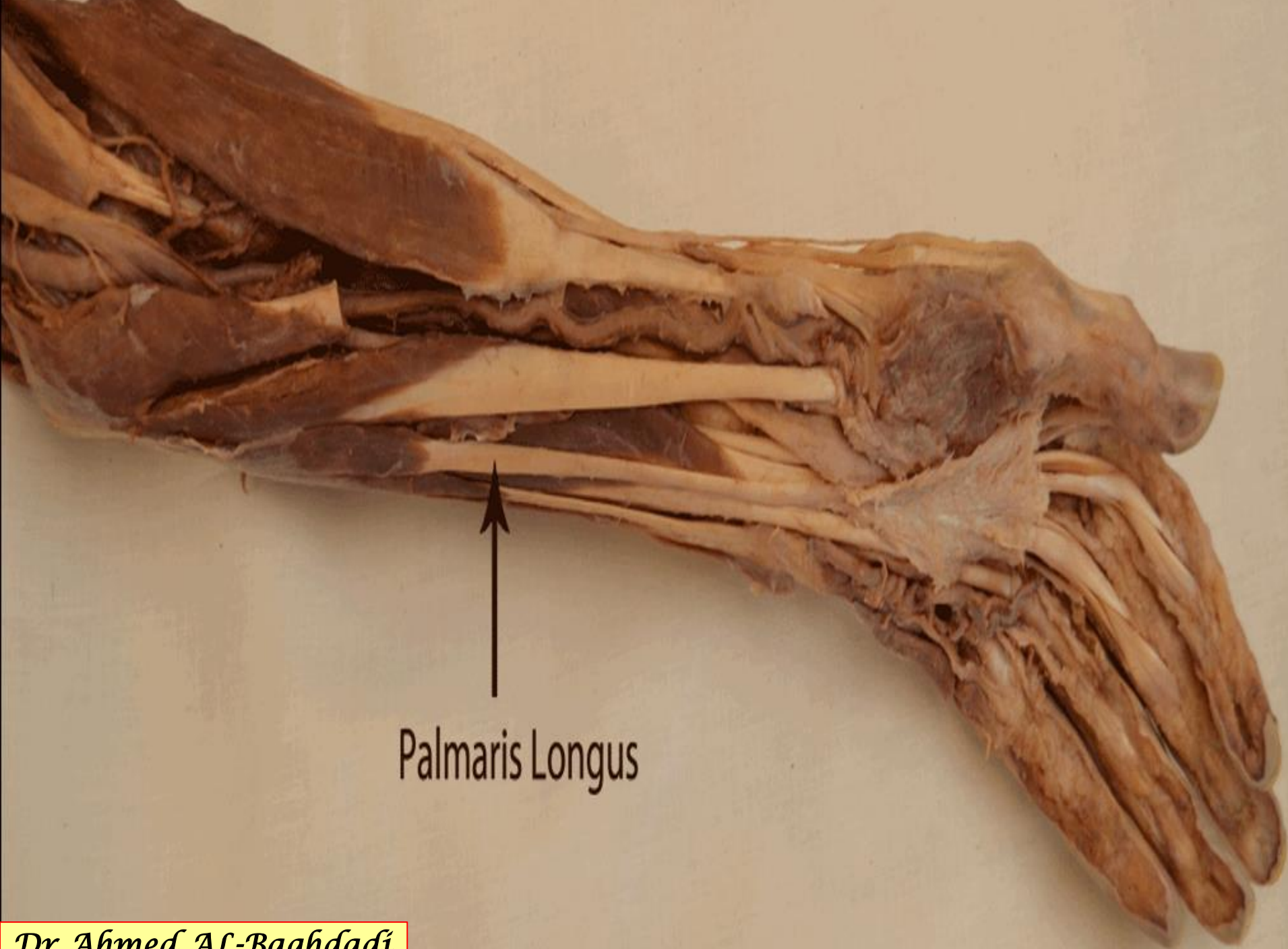
- 1=PRONATER TERES**
- 2=BRACHIORADIALIS**
- 3=BRACHIALIS**
- 4=SUPINATOR**
- 5=BICEPES TENDON**

**Cubital Fossa**

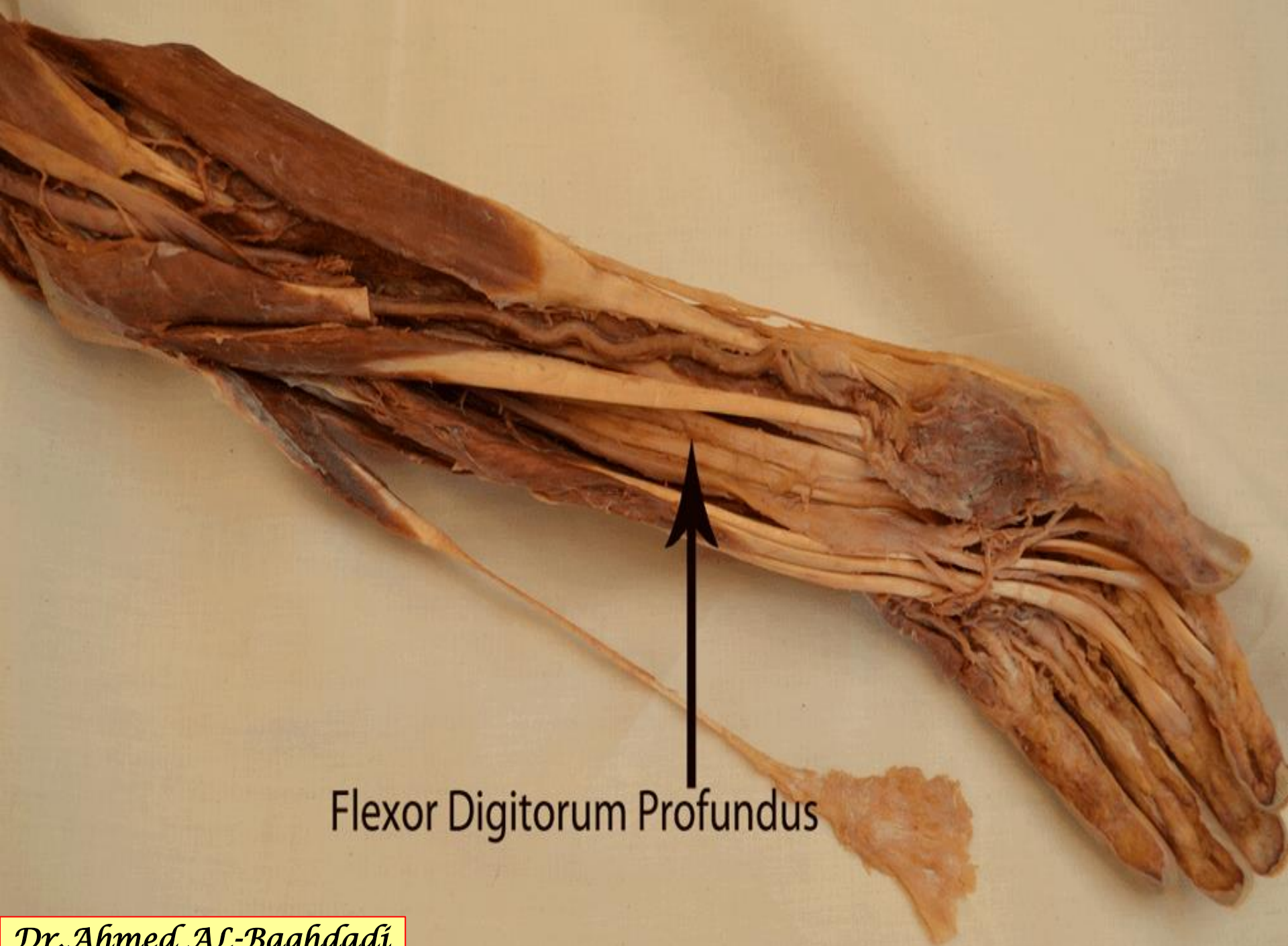






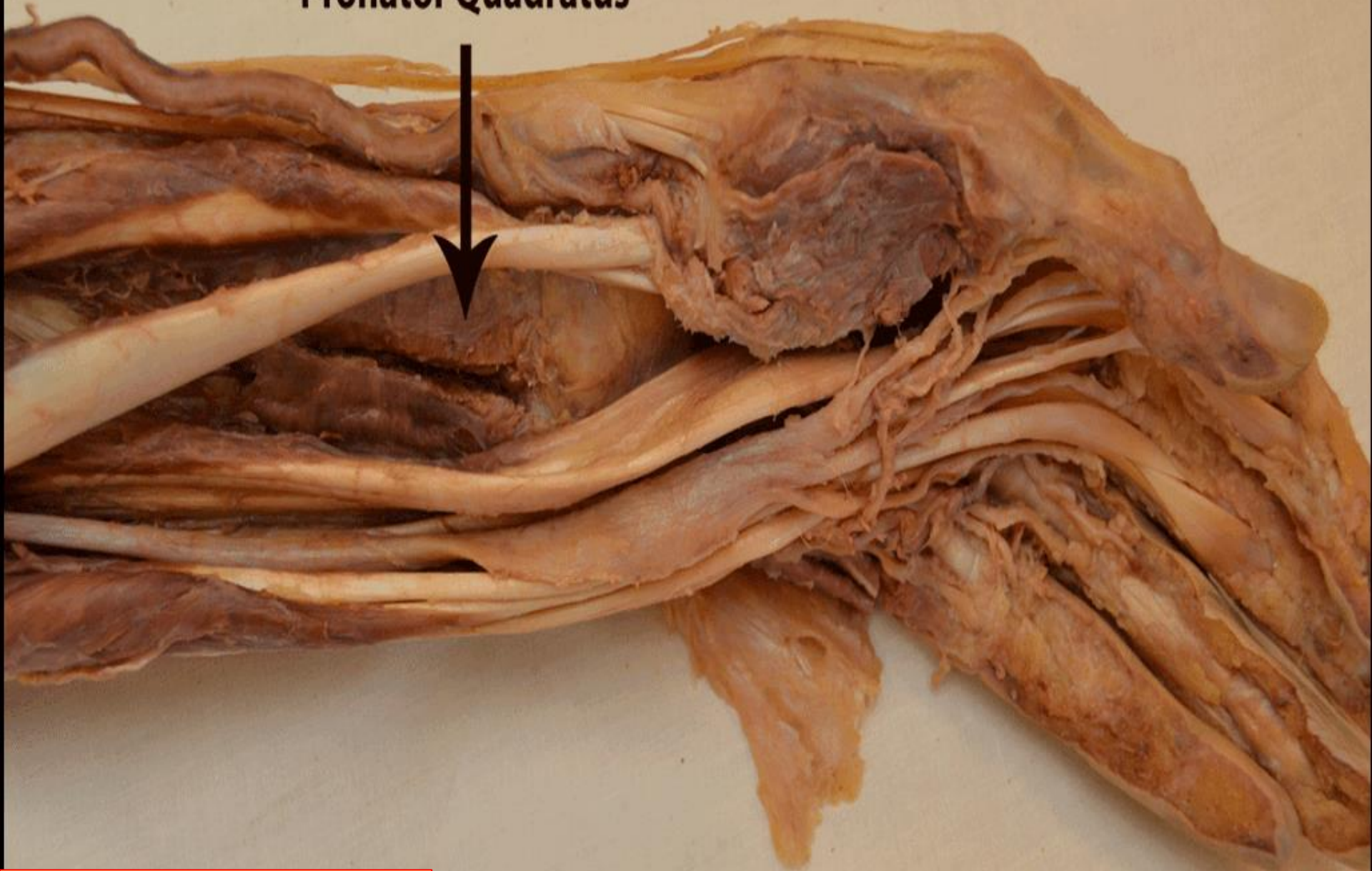


Palmaris Longus



Flexor Digitorum Profundus

**Pronator Quadratus**



**Extensor Carpi Radialis Brevis**

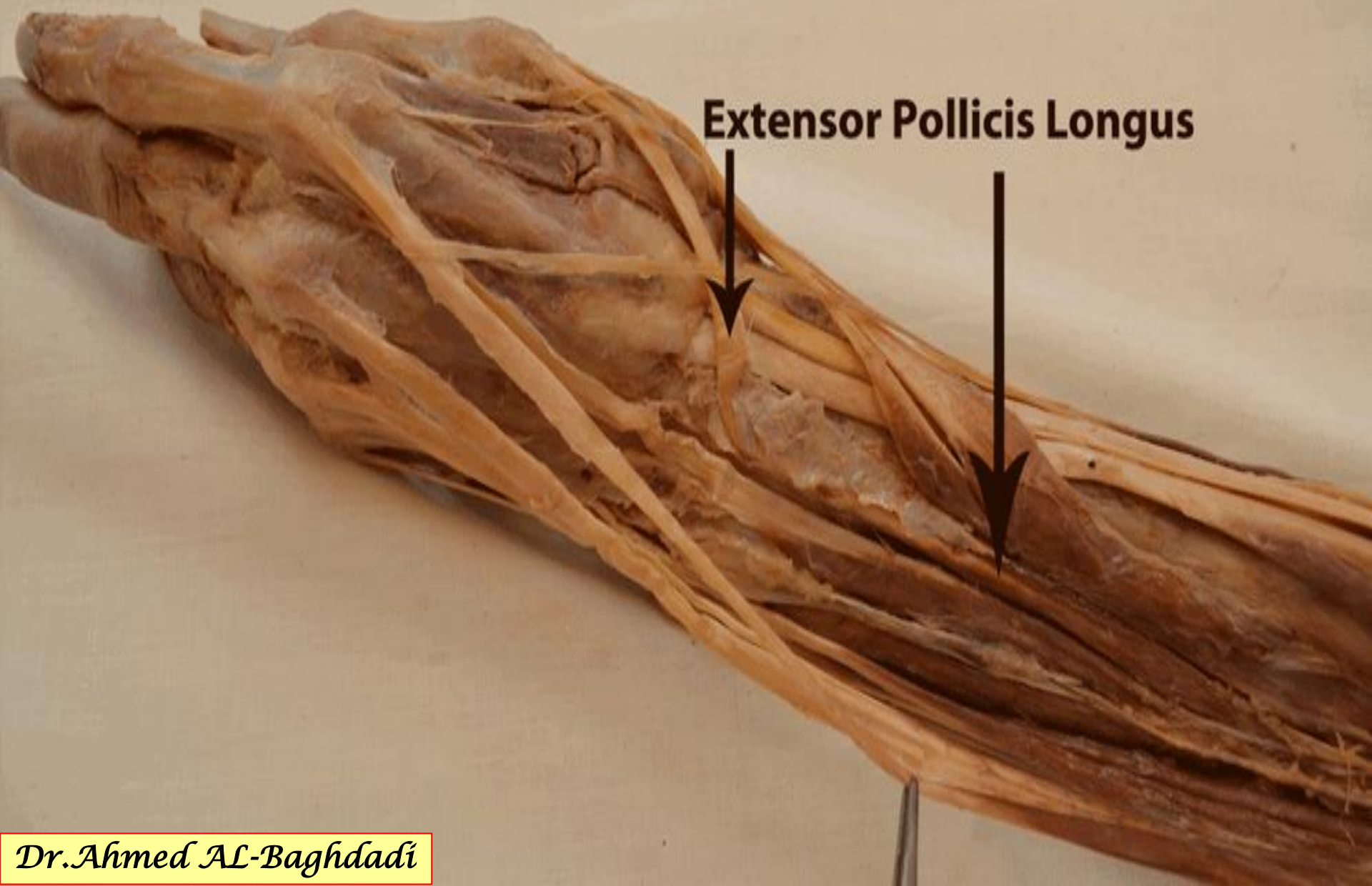




Anconeus



Supinator



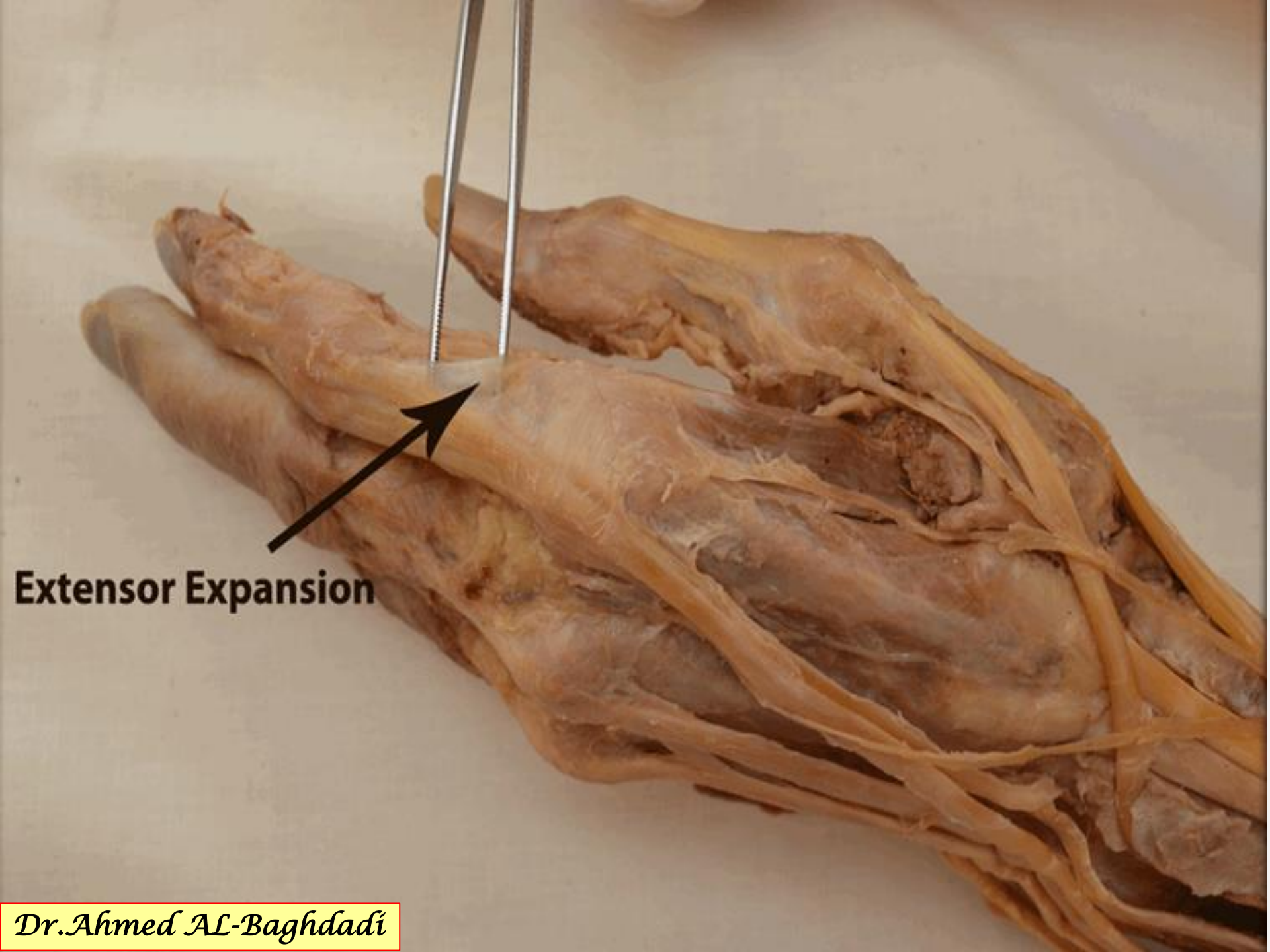
**Extensor Pollicis Longus**



Anatomical Snuffbox

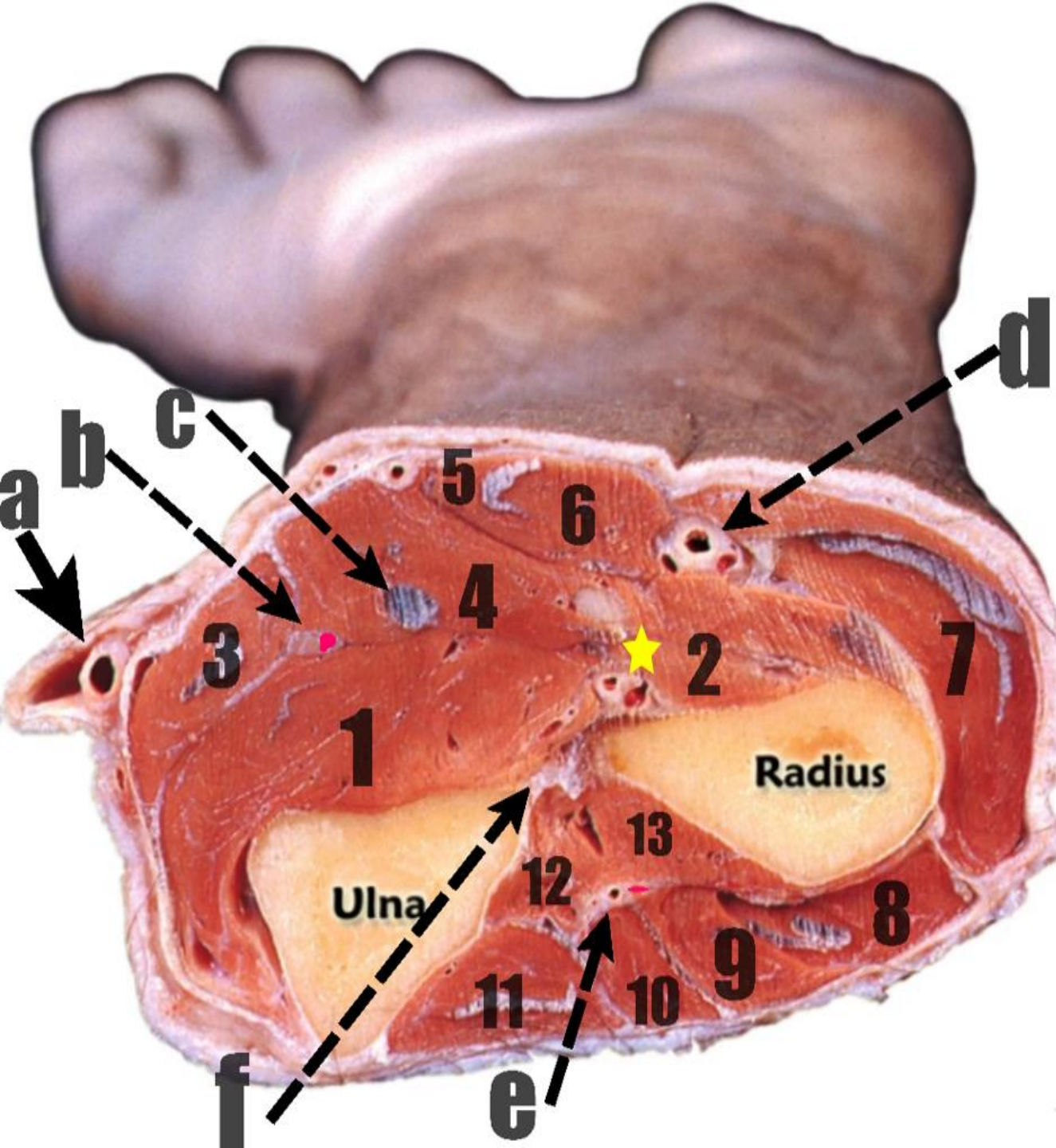


**Intertendinous Connections**



**Extensor Expansion**

## Axial section of forearm



a= Basilic vein

b= Ulnar nerve & artery

c= Median nerve

d= Radial nerve & artery

e= Post. interosseous nerve & artery

f= interosseous membrane

1= FDP

2=FHL

3=FCU

4=FDS

5= Palmaris longus

6FCR

7=Brachioradialis

8=ECRL

9= ED

10=EDM

11=ECU

12=EPL

13=ABD. HL

STARR= Ant. Interosseous nerve & artery

# RELATIONSHIP OF STRUCTURES ON THE FRONT OF THE WRIST

**structures lying in front of the conventional wrist from lateral to medial side are:**

1. Radial artery.
2. Tendon of flexor carpi radialis (FCR).
3. Median nerve
4. Tendon of palmaris longus.
5. Flexor digitorum superficialis.
6. Ulnar artery.
7. Ulnar nerve.
8. Tendon of flexor carpi ulnaris.

